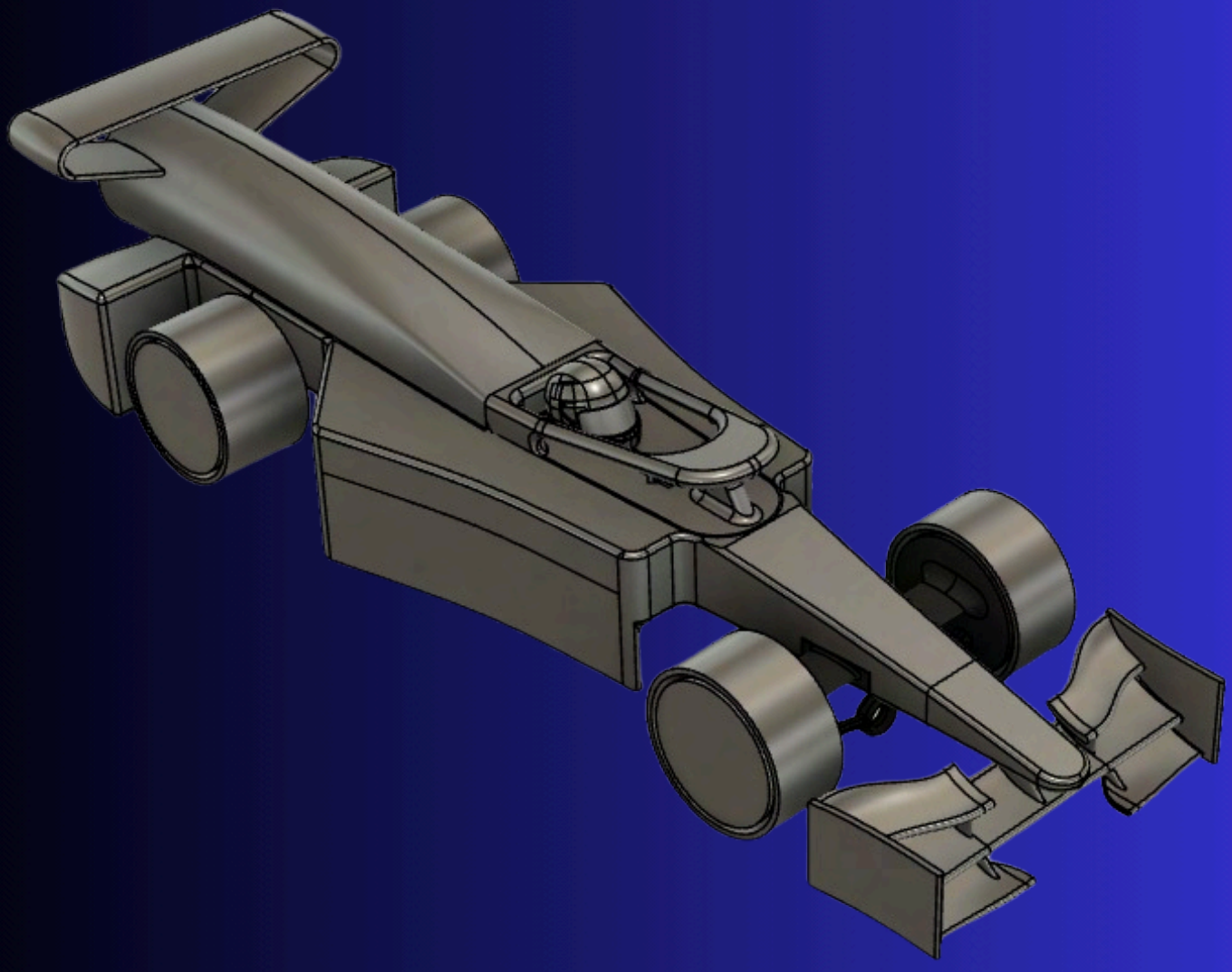
INFINITY RACING

ESTD. 2022

AERODNYAMICS SIMPLIFIED

THE ULTIMATE GUIDE FOR A FAST CAR



LAKSHYA HIRAWAT & ARYAMAN KEJRIWAL

POWERED BY: Ansys

INFINITY RACING - F1 IN SCHOOLS

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CHAPTER 1:

PREFACE

Infinity Racing, a group of six curious teenagers, participated in the F1 In Schools World Finals Competition in 2023 and 2024. We were newcomers in the contest and started our journey on the 24th of March 2022. Each of us reside in Mumbai, India and are from different schools. Two of our members, Lakshya Hirawat and Aryaman Kejriwal wrote this book to help you understand key concepts on aerodynamics used in F1 cars.

Our team consists of young teenagers who are thirsty for knowledge and are extremely interested in STEM (Science, Technology, Engineering Mathematics) as well as Formula 1. Hence, this competition was a complete must-do for us. Our members are:

Lakshya Hirawat (Team Manager, Graphic Designer), Aryaman Kejriwal (Team Manager, Design Engineer), Nathan Doshi (Sponsorship Manager, Manufacturing Engineer), Raghav Sharma (Sponsorship Manager, Graphic Designer), Yuvraj Ruparel (Design Engineer, Manufacturing Engineer) and Rajvir Sehgal (Resource Manager, Graphic Designer).

WHAT IS F1 IN SCHOOLS?

F1 In Schools is a global STEM competition in which students from countries all over the world compete to ultimately win the World Finals and be crowned the 'World Champions', a goal that every team has set. Each team consists of 3-6 members, each assigned a role which in a way symbolises the working of an actual F1 team. Each team must design a car using the official F1 model block using either CAM or CAD software, showcasing their skill in STEM.

Aside from designing the car the team also has to prepare various documents about the team's journey and design of the car in the form of two portfolios - the Project Management & Enterprise Portfolio as well as the Design & Engineering Portfolio. Every team must also follow a standard set of regulations in order to deem their car legal to race.

There is not only the Technical Regulations but also the Competition Regulations.This competition helps to enhance every member's designing skills as well as their ability to work in an enterprise.

OUR ACHIEVEMENTS

We take immense pride in all of our team's achievements throughout our F1 In Schools endeavour. Infinity Racing has gone through two tough stages of the competition: The India Regional Qualifiers and the India National Finals. While competing against 55 teams at the Regionals which was certainly a challenge, Infinity Racing bagged two awards, The Best Newcomer Award and The Best Project Management & Enterprise Portfolio Award. Along with these prizes, Infinity Racing was crowned the CHAMPIONS of the India Regional Qualifiers.

This started our journey for the India National Finals 2022. Even though our team started our work with confidence after just being the Champions, we were extremely nervous for the Nationals. We were told that among 50 teams from the Nationals only the top 3 teams would qualify for the World Finals 2023 in order to represent their country at the highest level of the competition. The fear and nervousness in our minds did not stop us from working as the National Finals came closer.

At the National Finals, Infinity Racing won a total of 3 awards as well as being nominated for a total of 7 awards. We won the People's Choice Award, The Best National Collaboration Award and lastly the Best Scrutineering Award. As we aspired to come in the top 3, the results were finally announced. Infinity Racing stood 3rd (2nd Runner's Up) at the Nationals and had qualified for the World Finals 2023.

We cannot wait for what the World Finals has in store for us. However, we do predict to see saffron, white and green colours on the podium at the World Finals.

Wish us the best of luck and support us as we set sail for our most ambitious campaign yet: The Formula 1 In Schools World Finals 2023.

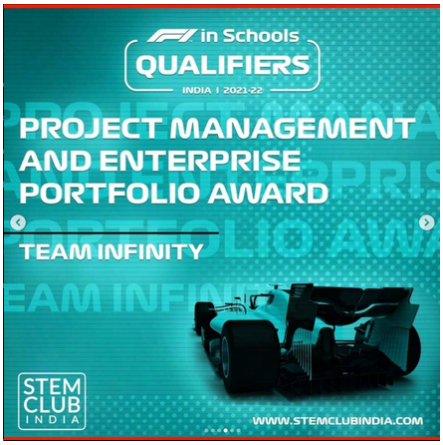
OUR ACHIEVEMENTS

Take a glimpse at some of our prestigious awards through our F1 In Schools endeavour.

India Regional Qualifiers 2022:







National Finals India 2022:











CHAPTER 2:

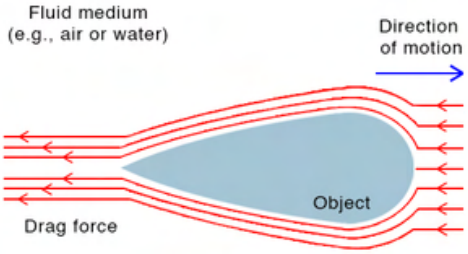
F1 AERODYNAMICS

Drag Force:

Drag is the aerodynamic force that is opposite to the velocity of an object moving through air. Its size is proportional to the speed differential between the air and the solid object.

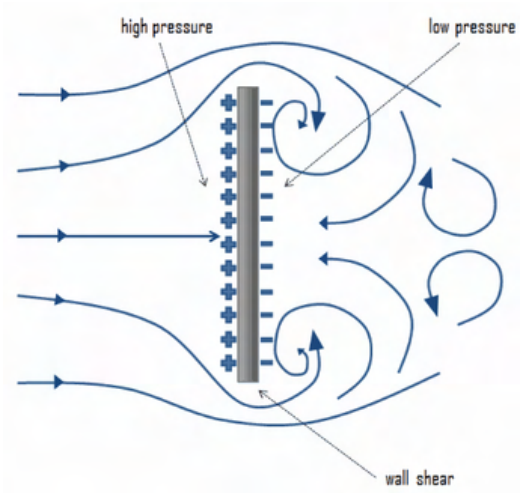
There are three types of drag force experienced by F1 in Schools cars

- Friction Drag
- Form Drag
- Interference Drag



| Fig 1.1 Image of airflow on teardrop shape [Effective for reducing drag]

Friction drag which is the result of the friction of the solid molecules against air molecules in their neighbourhood. Friction and its drag depend on both the fluid and the solid properties. A smooth surface of the solid for example produces less skin friction compared to a rough one.



| Fig 1.2 Image of air moving around a body

Another type of drag is a form of resistance from the air against the solid moving object. This form of drag is dependent on the particular shape of a wing, and is therefore called form drag. As air flows around a body, the local velocity and pressure are changed, & effectively creating a force.

Interference drag or induced drag on the other hand is the result of vortices that are generated behind the solid object. Due to the change of direction of air around the wing, a vortex is created where the airflow meets unchanged, straight flow. The size of the vortex, and thereby its drag strength increases with an increasing angle of attack of the aerofoil.

There are two more types of drag which are mostly irrelevant and negligible in race-cars, wave drag and ram drag.

The amount of drag that a certain object generates in an airflow is quantified in a drag coefficient. This coefficient expresses the ratio of the drag force to the force produced by the dynamic pressure times the area.

The formula to calculate the drag coefficient is:

$$F_D = \frac{1}{2}\rho v^2 C_D A$$

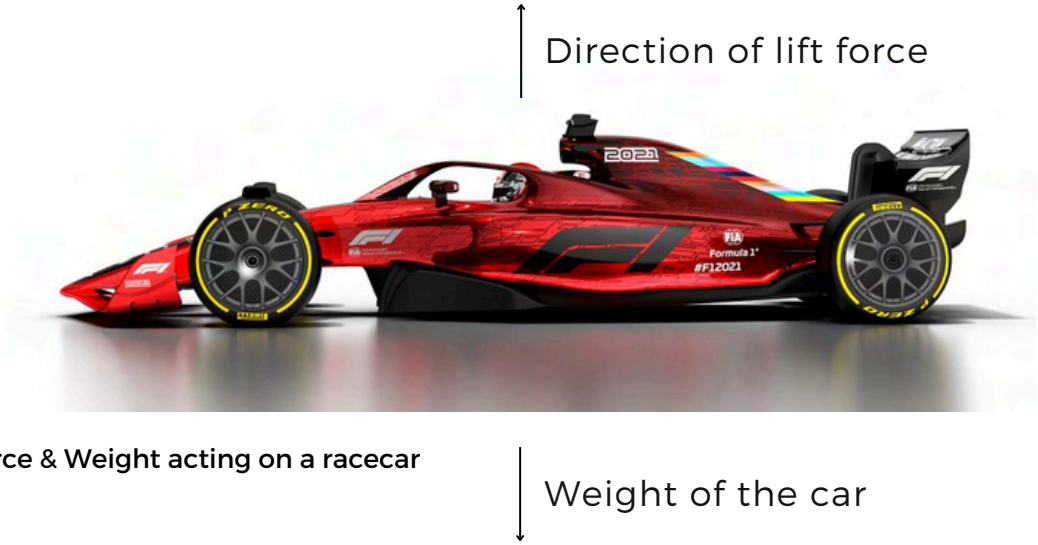
| Fig 1.3 Formula to calculate Drag Coefficient

where Fd is drag force, p is the density of the fluid, v is the flow velocity relative to the object, Cd is the drag coefficient, and A is the reference area.

Lift and Downforce:

Lift force directly opposes the weight of a race-car, pushing it up. It is created by every part of the car, and increases when there is greater air resisstance on the vehicle.

Here is a diagram to show how the lift force acts on the car.



| Fig 1.4 Lift force & Weight acting on a racecar

Lift occurs when a moving flow of gas is turned by a solid object. The flow is turned in one direction, and the lift is generated in the opposite direction, according to Newton's Third Law of action and reaction. Because air is a gas and the molecules are free to move about, any solid surface can deflect a flow.

When greater lift is produced, the wheels move slightly are pushed to the ground. Hence, the wheels do not get good grip off the track, reducing speed of rotations and magnitude of forward force due to rotations, slowing down the vehicle significantly.

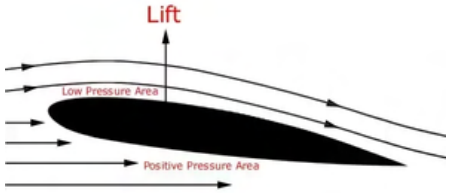
When greater lift is produced, the wheels move slightly are pushed to the ground. Hence, the wheels do not get good grip off the track, reducing speed of rotations and magnitude of forward force due to rotations, slowing down the vehicle significantly.

The amount of lift that a certain object generates in an airflow is quantified in a lift coefficient.

It is calculated using the Lift Equation

$$L = C_l\left(\frac{1}{2}\rho v^2\right)A = C_l q A$$

Where $q = \frac{1}{2}\rho v^2$



where L is the lift force, Cl is the lift coefficient, p is the density of the fluid, v is the flow velocity relative to the object, A is the wing area, and q is the dynamic pressure.

Downforce, or negative lift, pushes the car on to the track. Downforce has to be balanced between front and rear, left and right.

We can easily achieve the balance between left and right by simple symmetry. Front and rear is a different thing. Flow in the front greatly affects flow in the back of the car, and vice versa.

Many people confuse downforce with weight, when they are not the same thing, because weight force of an object due to gravity, acting on the center of mass, while downforce is produced by the whole car.

The following diagram shows the downforce produced by each part of a F1 race-car.

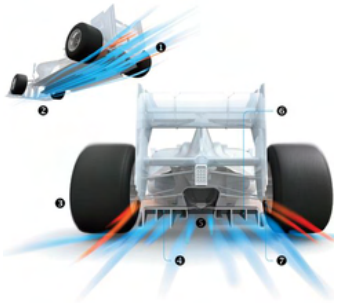


When downforce is created, the car stays firmly on the track, and the wheels get good grip, which increases the speed of rotation and the magnitude of forward force created by each rotation, speeding up the car.

While these simplified versions are the basics of lift and downforce generation, the reality can hardly be simplified and is a complex study, requiring high power computer systems. For a gas, we have to simultaneously conserve the mass, momentum, and energy in the flow. Hence, a change in the velocity of a gas in one direction results in a change in the velocity of the gas in a direction perpendicular to the original change.

The simultaneous conservation of mass, momentum, and energy of a fluid (while neglecting the effects of air viscosity) are called the Euler Equations after Leonard Euler. Several computer algorithms are based on these equations to make an approximation of the real situation.

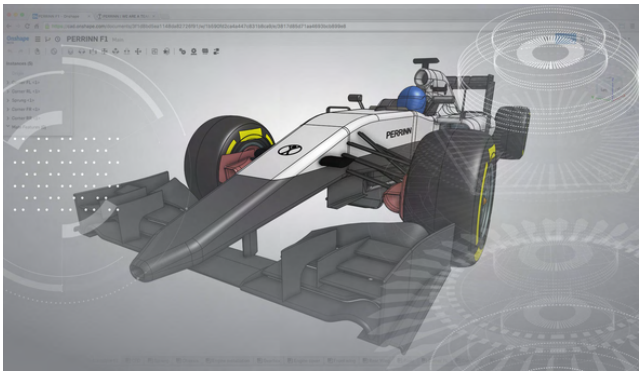
There are many parts in F1 cars used to create downforce on the car, such as vortex generators, diffusers, spoilers and more. Spoilers are used to reduce the gap between the ground and the vehicle so there is less air goes under, reducing lift.

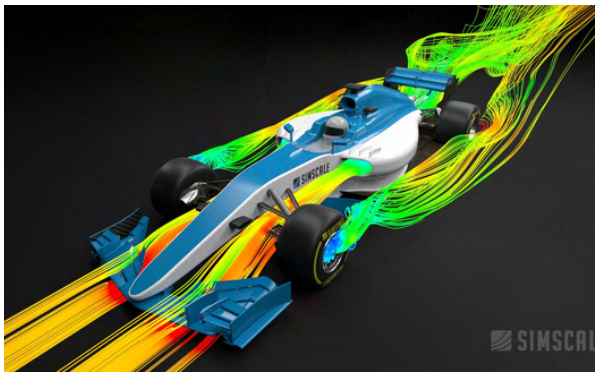




Too much downforce is not beneficial for optimum speed, as more downforce may create more friction, reducing the speed of the car on the track.

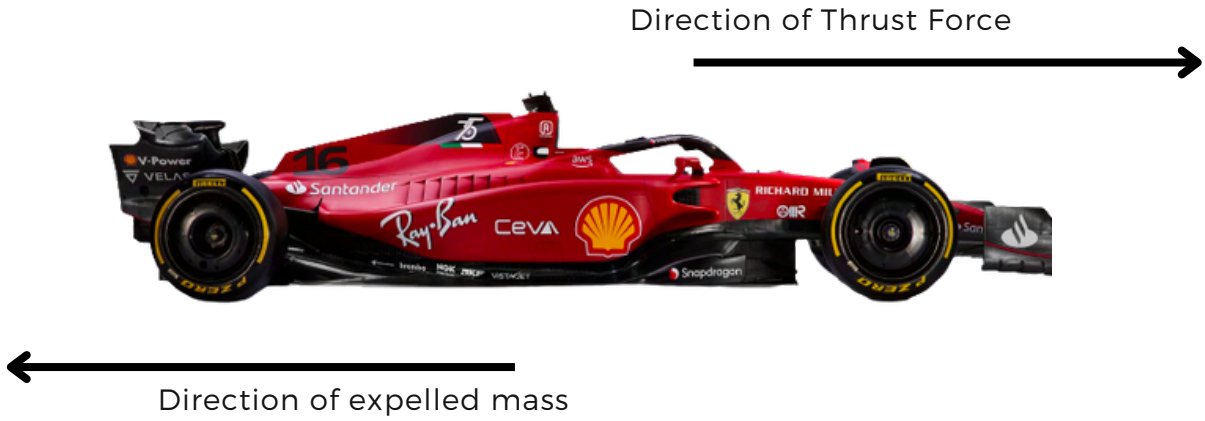
Because of the complexity, today's Formula One cars are designed with CFD (computational fluid dynamics) and CAD (computer aided design) that allows engineers to design a car, and immediately simulate the airflow around it, incorporating environmental parameters like traction, wind speed and direction, and much more.





Thrust Force:

Thrust is the reaction force which is described by Newton's third law. According to Newton's law, when a system expels mass in one direction, the accelerated mass will cause a force of equal magnitude but in the opposite direction to the direction in which the mass is expelled. This force applied in the opposite direction is know as thrust force.



In F1, the concept of Thrust force comes in play for the car to accelerate. The engine of the car uses the fuel to expel masses of air from the rear end of the car and this expelled mass of air results in a force in the opposite direction. It is this thrust force which causes the object to move.

In the F1 in schools competition the fuel for each car is the same as in this competition the CO_2 cartridge chamber serves as the fuel. It is essential for the teams to conserve this energy given out by the CO_2 cartridge. Most of this gas is expelled initially from the cartridge and most of the energy goes to waste.

To make the most efficient use of the energy given out by the CO2 chamber, one of the most effective methods is to design a system in the car which would conserve the energy given out by the CO2 cartridge and release the energy at regular intervals. This type of system is known as an LERS device. By using this device, the energy system of the car is more efficient.

Another way of increasing the thrust force of the car, is by incorporating another of Newton's laws which states :

$$F = ma$$

wherein; F - Force, m - mass and a - acceleration

The force refers to the thrust force in our case. Since the thrust force would be relatively the same for all the cars unless the teams have incorporated an LERS device system into their car design. Hence to increase the acceleration of the car it was essential to decrease the other variable which is the mass of the car.

There are several ways in which a team can decrease their car's mass. Two of the most effective ones are :

1. To make the body of the car hollow
2. Use relatively less weight 3D printing material for the manufacturing of the cars.

However from these methods, it is recommended to use the first option as many of light weight 3D printing material are not very strong and are not able to withstand the pressure on it. Another important precaution to keep in mind while making the body of the car hollow is to make sure that any part of the car is not too thin such that there would be a tendency for it to break during its manufacturing or racing. One of the easy ways to make the body of the car hollow is use the shell tool or cut extrude the body.

Making these changes to the car would greatly enhance the performance of the car. Reducing the mass of the car is the most effective way to increase the acceleration of the car and it is crucial to pay attention to this while making the car. To design a winning car it is essential that its mass is less than 52 grams. It is also necessary to ensure that not too many layers of paint are added to the car during its manufacturing which would add to the mass of the car.

ACTIVITY - RESEARCH TIME



Research on different ways you can reduce the mass of your car. We recommend you'll to draw diagrams and add pictures to get a better understanding of how you'll can reduce the mass of your car. Your team can also research on other ways of increasing acceleration.

CHAPTER 3:

SCIENTIFIC CONCEPTS

Magnus Effect:

The Magnus effect is a scientific concept is a phenomenon which is observed when a spinning object is moving through a fluid. This occurs when there is a relative motion between the spinning body and the fluid. It is responsible for the 'curve' of a served tennis ball or driven golf ball and affects the trajectory of a spinning artillery shell.

A spinning object moves through its straight path because of the pressure differences that develop in a fluid as a result of velocity changes induced by the spinning object.





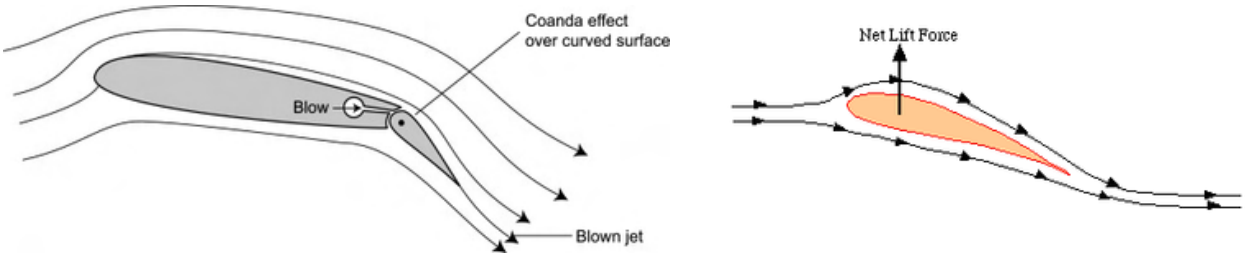
| Fig 2.1 Lift force & Weight acting on a racecar

The Magnus effect can be interpreted as a manifestation of Bernoulli's Theorem, fluid pressure decreases at points where the fluid speed is increased.

Coanda Effect:

The Coanda effect in simple words, is the tendency of a fluid to stay in contact with a convex surface. The effect of this concept has significant impacts in the aerodynamics related vehicles. It is due to the Coanda effect that the aeroplanes stay in the air as the air around sticks to the wings and body of the aeroplane.

To understand the effect of this concept, let us consider an example. When water flows off a sloping roof it sticks to surface of a gently sloping gutter. A similar idea can be used in our cars to redirect the airflow around the car so that it does not come in contact with any other car part.



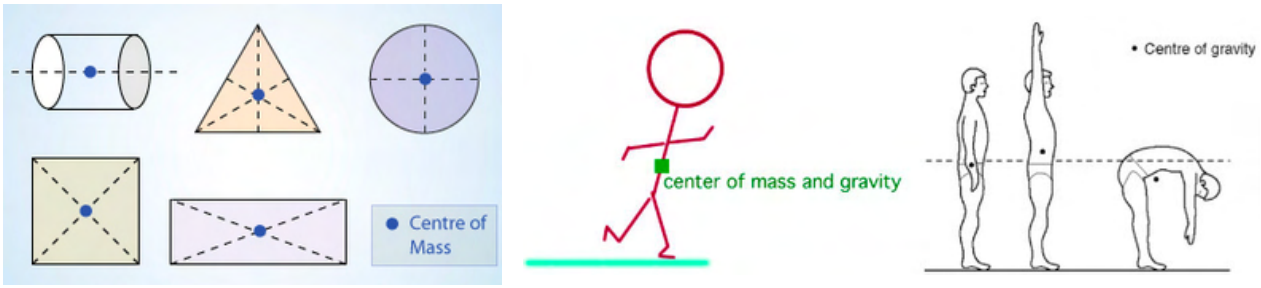
| Fig 2.2 Effect on the airflow due to the Coanda effect

Another result of this effect is that it causes lift which needs to be taken note of when designing a car. Hence increasing the need to have downforce due to the lift created making it essential for F1 cars.

In F1 in schools, Coanda effect can have a significant impact when designing the wings and sidepods. If they are designed in such a way that they can redirect the airflow above the wheels and body it will not prevent the rotation of the wheels and also reduce the air resistance faced by the car. Similar effects when designing the sidepods. So to maximise the performance of the car it is essential to take note of this phenomena.

Centre of Mass :

Centre of mass goes about by many names - centre of gravity and even centre of weight. They all mean the same thing. Centre of mass is basically a point in a body where all the mass of a body is supposed to be concentrated. The centre of mass is usually located at the centre of a body.



| Fig 2.2 Effect on the airflow due to the Coanda effect

The centre of gravity is perhaps the single most critical area of any racing car weight consideration. It must be as low as possible to the ground, and this is not only because the weight act through it, but also because all accelerative forces of acceleration, braking and cornering work trough it.

Any force that acts through the CoG has no tendency to make the car rotate. For a racing car, the CoG is located between the front and rear wheels and as close to the ground as physically possible.

It is good to have the static weight distribution tuned in way that the centre of gravity is positioned closer to the rear of the car, as this puts more weight on the rear wheels and provides more traction,

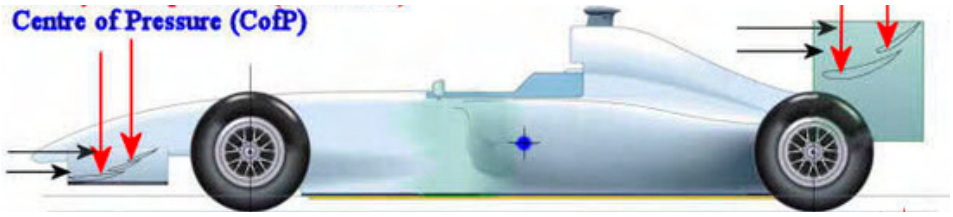
Centre of Mass and Stability :

The location of the centre of mass in a car can have serious impacts of on the performance of your car. There are two significant impact of the centre of mass on the stability of the car. Stability is the likelihood of the car to not topple over.

The two significant impacts are :

1. *The height of the location of the centre of mass*

   You might have noticed that most of the F1 cars have bodies which are located very close to the ground. This plays a tremendous role in keeping the car stable and not toppling over. The lower the centre of mass the more stable the car is and a higher tendency for the car to stay in contact with ground and the wheels not come off the ground, hence increasing the efficiency of the car.



| Fig 2.2 Effect on the airflow due to the Coanda effect

2. *The centre of mass being in the middle from the right weel and left*

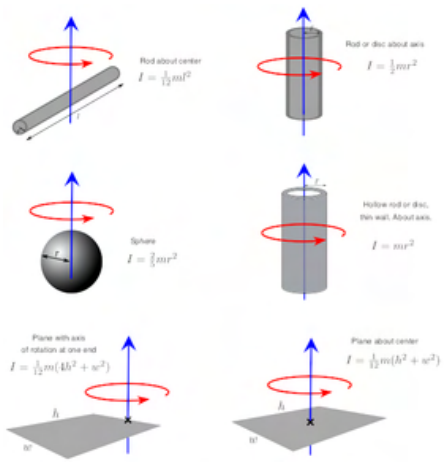
   If any of the wheels of the car leave the surface it is important that the centre of mass does not go onto the outer side of the wheel which would make the car unstable and cause it to topple over. This might not have a significant impact in F1 in schools because the tether line guide will not let the car topple over but it is important to know of when designing the car. However the first point is critical in F1 in schools.

Moment of mass and inertia :

Moment of inertia is a physical property of an object that describes its resistance to rotational motion about a particular axis. It is a measure of how the mass of an object is distributed around that axis of rotation. In simple terms, the moment of inertia quantifies the object's ability to resist changes in its rotational motion.

Effects of moment of inertia on a car:

a. Acceleration: A car with a larger moment of inertia will take longer to accelerate or decelerate in rotational motion. This means it will be slower to change direction or perform maneuvers such as turning or spinning.
b. Stability: The moment of inertia affects the car's stability during turns. A higher moment of inertia can make the car more stable since it resists changes in rotation. This can be beneficial for maintaining control during high-speed turns or in adverse driving conditions.



Newtons Laws of Motion :

The first law:

Newton's first law of motion, also known as the law of inertia, states that an object at rest will remain at rest, and an object in motion will continue moving in a straight line at a constant velocity, unless acted upon by an external force.

In our competition, F1 in schools, at all times during the race an external force would be acting on the car. At the start of the race there would be the CO2 chamber push which would act as the external factor. Then during the race there would be other factors like air resistance and rolling friction which would interfere with the movement of the car.

From this law, we can understand that the car would not have the same speed throughout the race and the velocity will be variable according to the forces acting on it.

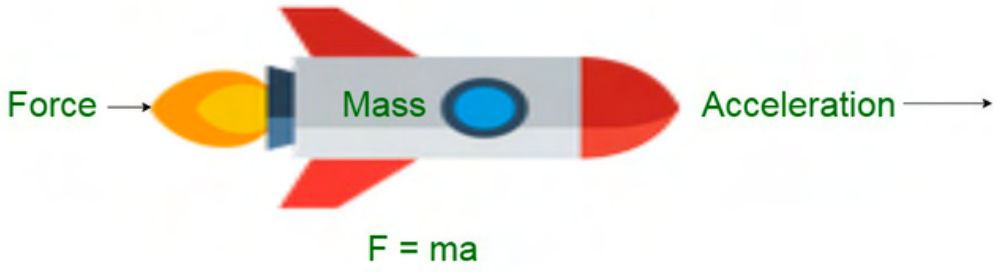
In summary, Newton's first law of motion, the law of inertia, influences various aspects of an F1 car's performance, including straight-line motion, braking, changing direction, cornering, and response to external forces. Understanding and applying this fundamental principle of physics is crucial for designing and optimizing the performance of F1 cars on the track.

Newtons Laws of Motion :

The second law:
Newton's second law of motion, also known as the law of acceleration, states that force is equal to the rate of change of momentum. This relates the net force acting on an object to its mass and acceleration. It can be stated mathematically as follows:

F = m * a

Where: F represents the net force acting on the object, m represents the mass of the object, and a represents the acceleration produced by the net force.



F = ma

In simpler terms, it explains how the motion of an object changes when a force is exerted upon it. According to this law, the greater the force applied to an object, the greater its acceleration will be, provided its mass remains constant. Conversely, if the mass of an object is increased, its acceleration will decrease for a given force.

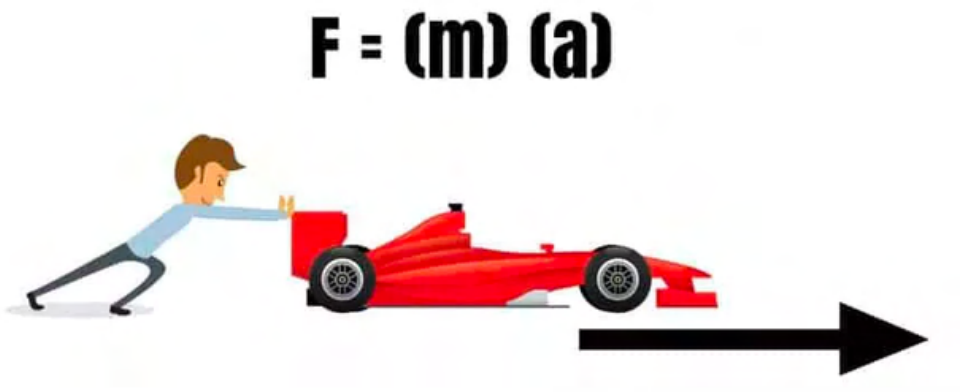
The understanding of the above equation is critical for setting a strong foundation for your aerodynamic concepts. This equation will play a key role in the design of your F1 in schools cars.

Newtons Laws of Motion :

The second law:

In the context of F1 cars, Newton's second law has several implications:

1. Accelleration
2. Force Balance



To get a better understanding of the impact of this law in F1 in schools cars. Let us consider an example of the image above :

In this picture, the man is applying some force F on the car. In our competition this force F is provided by the CO2 chamber canister and is the same for all teams. So the left hand side of the equation is the same for all teams. On the right hand side of the equation we have the mass and acceleration of the car. We want to maximise the acceleration of the car. In order to do so, we will have to decrease the mass of the car since the mass of the car is something we can control. We have to keep the mass of the car between 50-51 grams in our car to maximise the acceleration. Remember to keep in mind the mass that will be added during any manufacturing defect or painting and decals.

Newtons Laws of Motion :

The third law:

Newton's third law of motion, often referred to as the law of action and reaction, states that if an object exerts a force on another object, the second object exerts an equal and opposing force on the first object.

In practical terms, Newton's third law highlights the interconnected nature of forces. It states that whenever an object exerts a force on another object, the second object exerts an equal and opposite force back on the first object. This law helps us comprehend the cause-and-effect relationship between interacting objects. It reminds us that forces always occur in pairs and that they act on different objects simultaneously.

Newton's third law of motion has a significant impact on Formula 1 (F1) cars and plays a crucial role in their performance and dynamics. The law states that for every action, there is an equal and opposite reaction, and this principle directly applies to F1 cars in several ways.

Firstly, when an F1 car accelerates, the engine generates a force that propels the car forward. According to Newton's third law, as the engine exerts a force on the car in the forward direction, there is an equal and opposite reaction force exerted by the car on the engine. This reaction force pushes against the ground, providing the necessary traction for the car to move forward effectively.



Friction:

Friction is a force that resists the motion between two surfaces in contact. When two objects come into contact and one tries to slide or move against the other, friction acts in the opposite direction to impede the motion. It is caused by the irregularities and roughness present on the surfaces in contact.

There are several types of friction :

- Static Friction: This is the friction that acts on objects at rest, preventing them from starting to move. It can be thought of as the initial resistance to motion.

- Kinetic Friction: Also known as sliding friction, kinetic friction occurs when two surfaces are already in relative motion. It opposes the motion and acts to slow down or stop the objects.

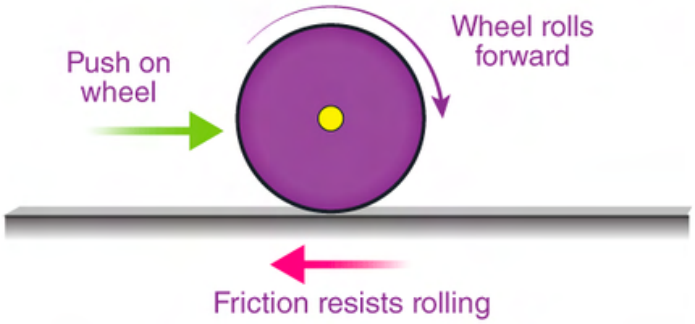
- Rolling Friction: Rolling friction is encountered when one object rolls over another, such as a ball rolling on the ground or a wheel on a road. It is generally less than kinetic friction and allows for smoother motion.

- Fluid Friction: Fluid friction, also called viscous friction, occurs when an object moves through a fluid like air or water. It opposes the motion and depends on factors like the shape of the object, the speed, and the properties of the fluid.

In F1 in schools the rolling friction and fluid friction will have a huge impact on the performance of your car. So will understand how to minimise each of these forces, which would increase the speed of our car.

Rolling Friction:

Rolling friction, also known as rolling resistance, is a type of friction that occurs when an object rolls over a surface. It is the resistance encountered by an object in motion that is in contact with a surface and rolling rather than sliding.



This rolling friction is faced mainly in two places:

- Between the wheel outer surface and the track
- The axle and the wheel

To minimise the effect of this friction it is important to reduce the impact of the friction in both these places.

The factors which affect the amount of rolling friction are :

- Nature of the surfaces: The materials and surface characteristics of both the rolling object and the surface it rolls on can affect the rolling friction. Rough surfaces tend to have higher rolling friction compared to smooth surfaces.

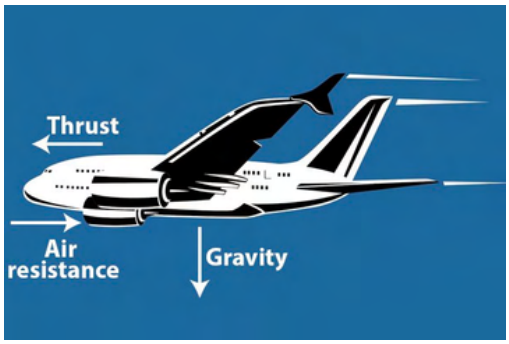
- Load or weight: The amount of weight or load pressing the object against the surface can influence rolling friction. Heavier objects typically experience higher rolling resistance.

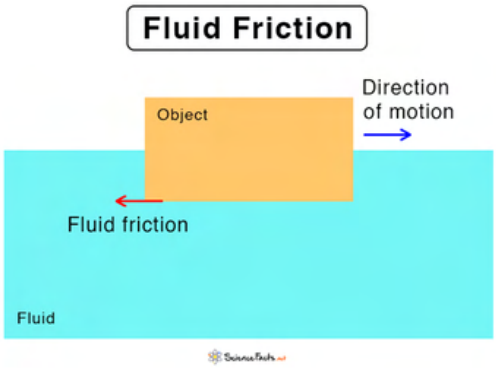
The most effective ways to reduce the rolling friction are :

- The appropriate use of bearings in the wheels
- Making sure the bearings are rotating smoothly
- The wheel and the having minimum weight
- The material used to manufacture the wheel does not produce

Fluid Friction:

Fluid friction, also known as viscous drag or simply drag, is the friction which a body faces when its set in motion through a fluid like air or water. In our case, the fluid is air and the friction is drag. The ways to minimise the drag faced by the car have been spoken about earlier.





ACTIVITY - Key Takeaways



After learning about the several different scientific concepts, make a note of the most important points you learnt, so you have easy access to them when you would be designing your car. Writing it down would also remember them better.

CHAPTER 4:

MAKING THE CAR

Software:

The two main softwares used in F1 in schools CAD are Autodesk's Fusion 360, and Solidworks.

Although Solidworks is the leading software for CAD, CAM and more, both Fusion 360 and Solidworks have their pros and cons.

Advantages of Fusion 360 compared to Solidworks:

- It can be run on both Windows and Mac Operating Systems
- Better cloud storage, can collaborate with people all across the globe
- Simpler user interface, for beginners
- 1 year free student license
- Lesser storage and RAM used - cheaper system needed



Advantages of Solidworks compared to Fusion 360:

- Much more advanced tools to create better designs in CAD
- Photorealistic rendering capabilities
- Greater user support
- Very accurate virtual simulations



After comparing the pros and cons, choose the most optimum software for you.

Body:

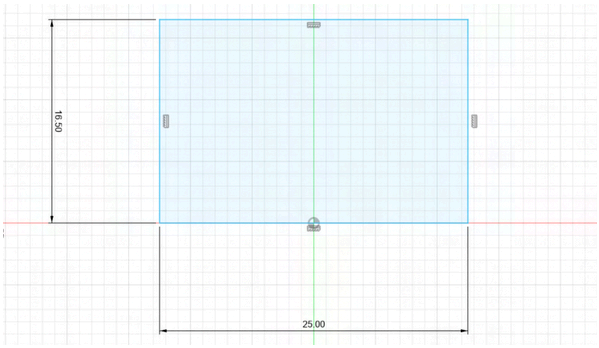
Drag is one of the main forces which slows down the car. Air that separates at the front of the car tries to move to its similar position at the rear end of the car. However, if a body of a car is not streamlined, the air tends to from turbulence at the rear and this occurrence is known as wake drag.

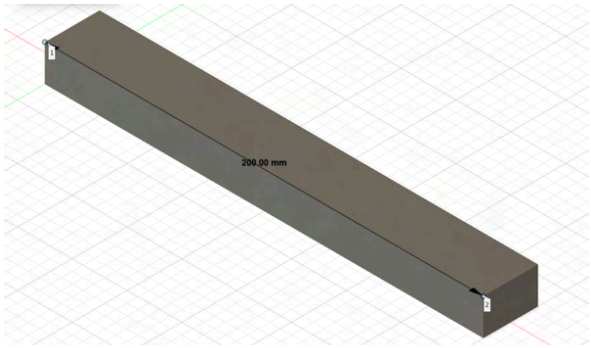
The air essentially is trying to fill a vacuum at the rear of the car and this vacuum is what slows down the car. This is why a car should contain streamlined chassis which would last the entire length of the car.

The streamlined effect of the nose cone has to be continued and maintained by the body, and with minimum surface area exposed.

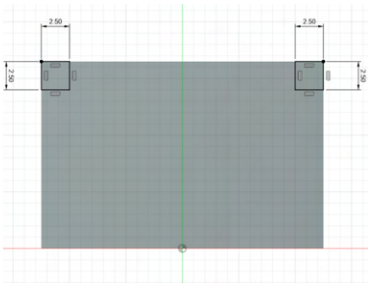
First we will start off with making the body:

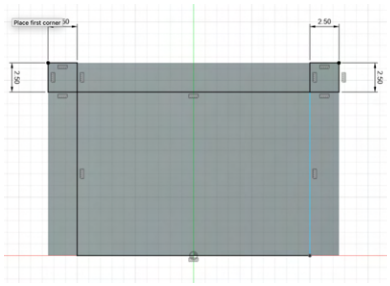
Step 1: Sketch a rectangle of length 25 mm and width 16.5, and extrude it by 200 mm.

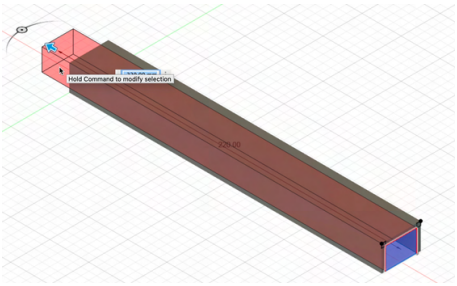




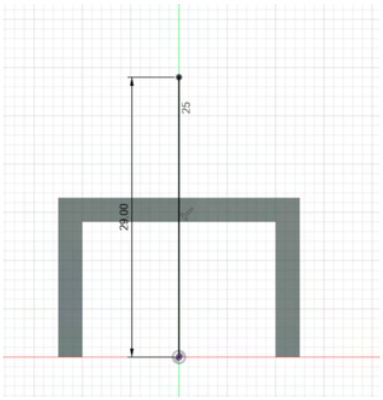
Step 2: In the front of the cuboid, sketch 2 square in the top left and top right corner of side 2.5 mm each. Create a rectangle touching the corner of both squares and extrude it throughout the body.

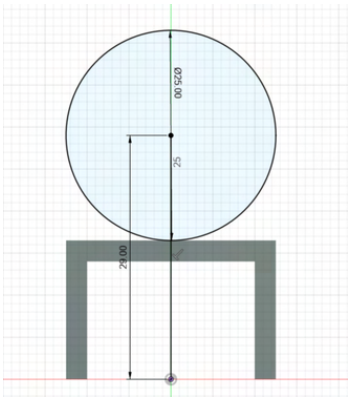


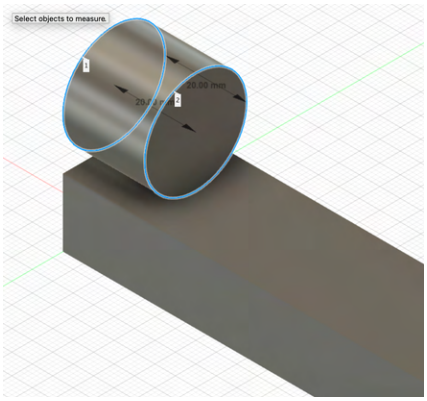




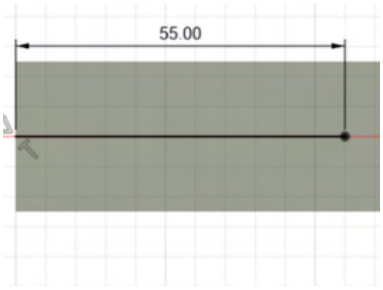
Step 3: At the back of the figure, sketch a line from the centre of the figure vertically 29 mm. Sketch a circle on the end of the line of diameter 25 mm, and extrude it 20mm.

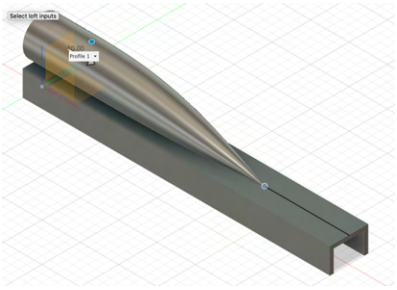


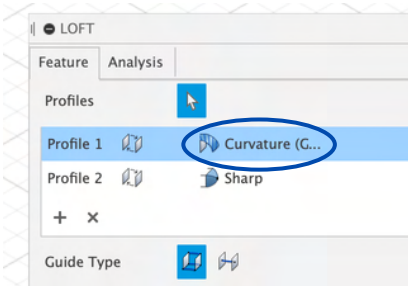




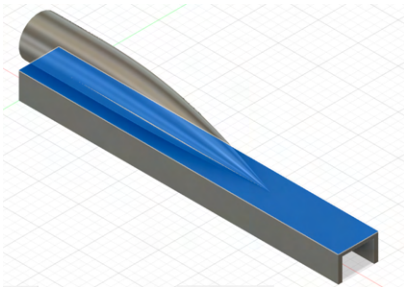
Step 4: Sketch a line on the top face of the figure, 55mm from the front. In the Create Menu, select Loft. First select the cylinder face and then the end point of the line. In the loft menu, click on "Connected" beside Profile 1 and select "Curvature".

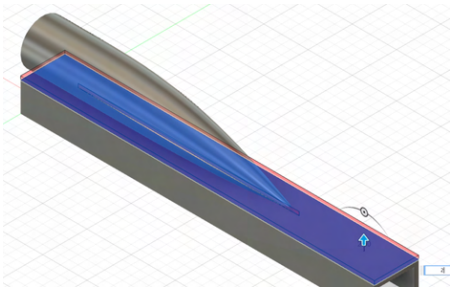


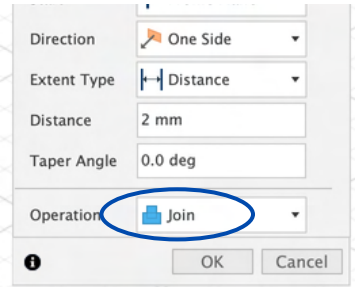




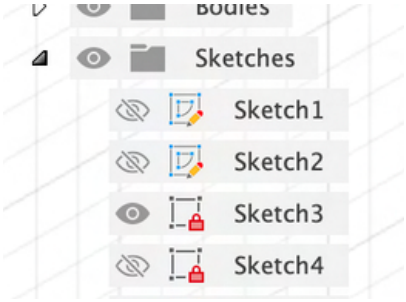
Step 5: Select the top face of the rectangular figure and extrude it by 2mm upwards. In the extrude menu, change the Operation setting from "Cut" to "Join"

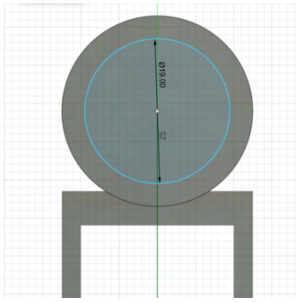


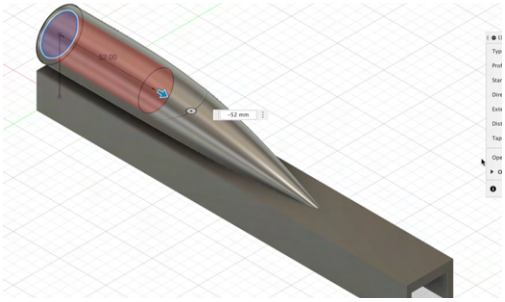




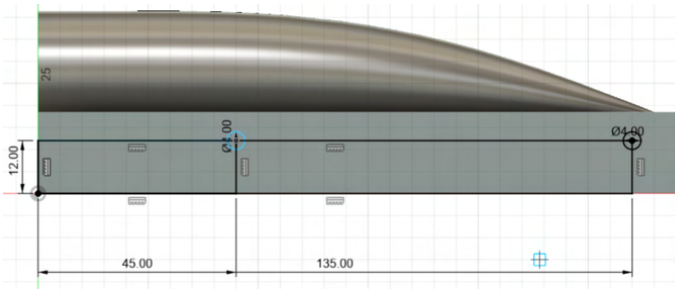
Step 6: Show Sketch 3, and create a new sketch on the same plane. Sketch a circle of diameter 19 mm at the end of the line and extrude it by -52 mm

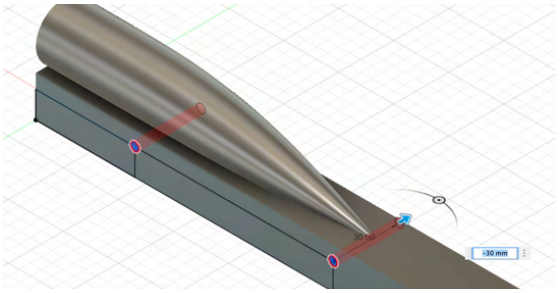






Step 7: On the side face, sketch two rectangles, from the bottom left corner, height 12mm for both and length 45mm and 135mm respectively. Sketch a circle on the top right corners of both rectangles of diameter 4 mm and extrude it by -30mm





The body is finished.

Nose cone:

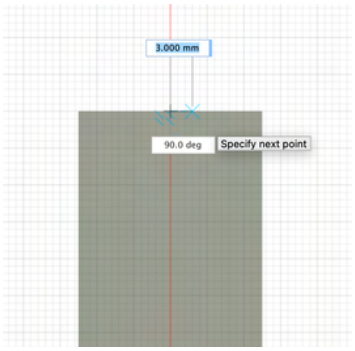
The nose cone is responsible for creating the aerodynamic effect, so in order for them to cut the air most efficiently, it has to be very streamlined.

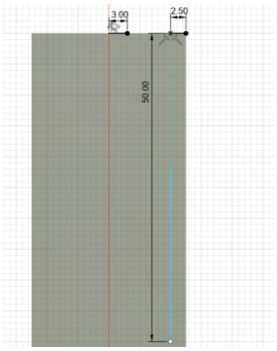
It is best to keep the sides of the nose cone 0° to the all x, y, and z axis and carve out the nose cone from inside the sides. This is because, we can then position the front wing and wheel axils well, so it isn't affected by the curved sides (demonstration later).

The nose cone must avoid friction drag, by keeping curved sides, edges and corners and must not push a lot of air under the car, to avoid lift.

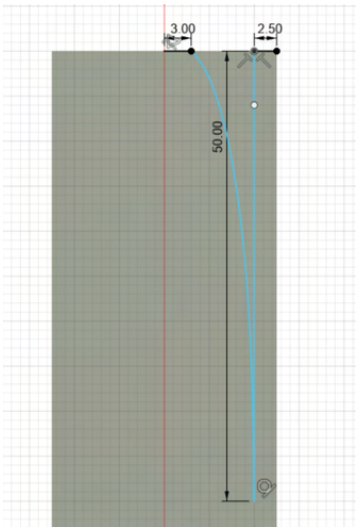
Now, lets start making the nose cone on the same design.

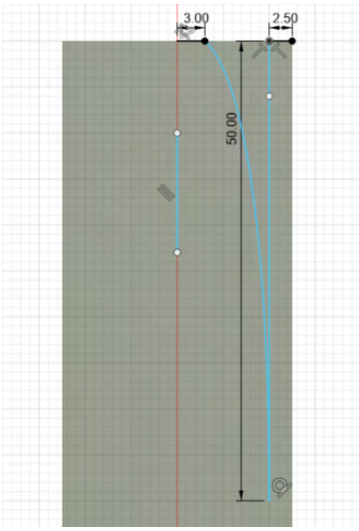
Step 1: Create a sketch on the top face, draw a line at the right side of the edge, of 3mm from the axis. From the top right corner, draw 2.5 mm line horizontally and a line 50 mm down.

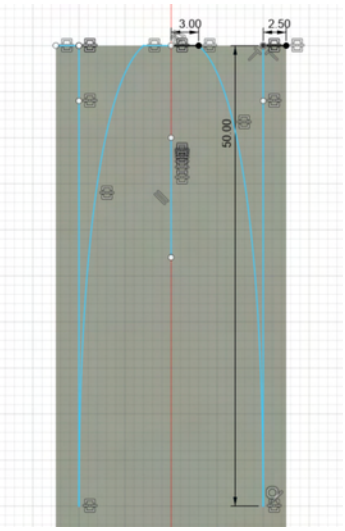




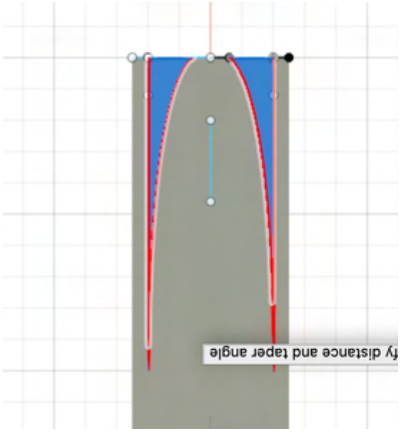
Step 2: Create a conic curve from the end of the 3mm line and end of the 50mm line, adjust the curve according to you. Sketch a vertical line on the axis. Using the mirror tool, mirror every line on the right side to the left, using the line as an axis.

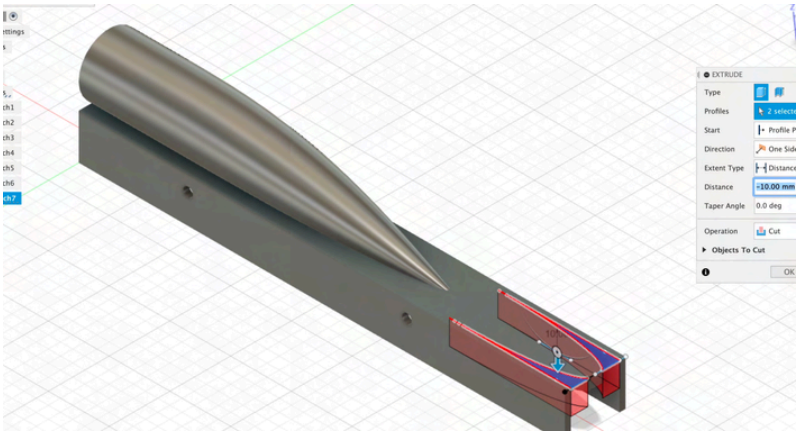




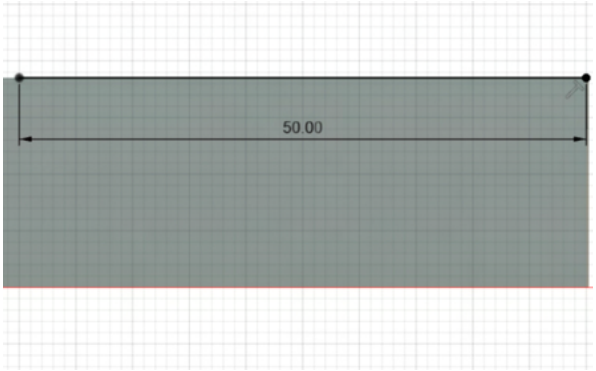


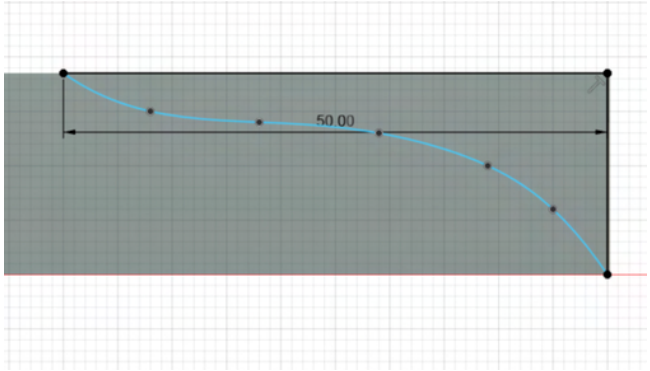
Step 3: Select the triangular shapes between the 50mm lines and the conic curves and extrude them by -10mm.



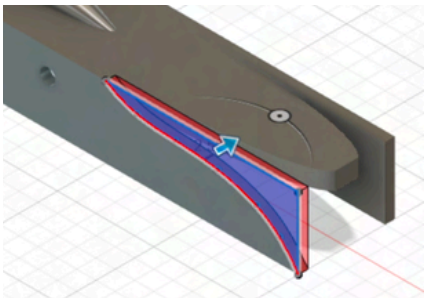


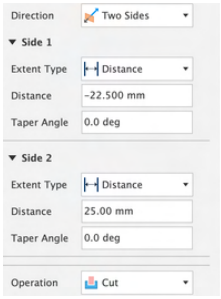
Step 4: On the right side face, create a sketch. Draw a 50mm line on the top from the top right corner Using fit point spline, create the following shape. Sketch a line from the bottom right to the top right corner.

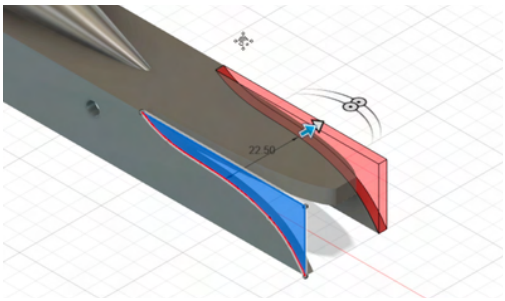




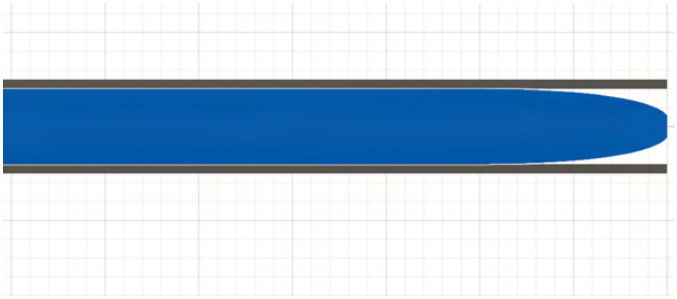
Step 5: Extrude the outer shape by -2.5mm. Reshow the sketch and extrude it. In direction select "Two sides" extrude Side 2 by 25 mmand Side 1 by -22.5 mm .

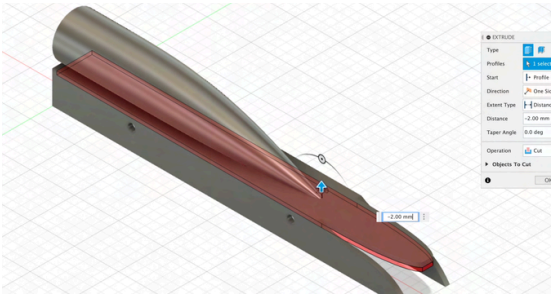




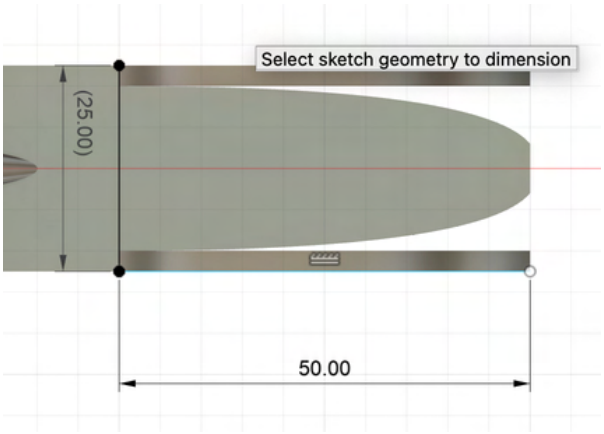


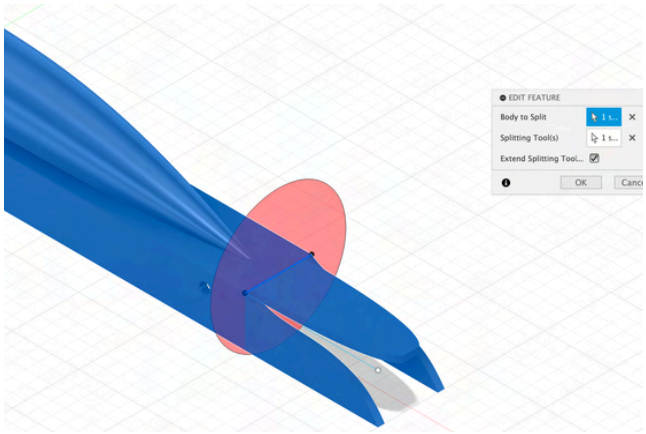
Step 6: Got to the bottom view. Extrude the face by -2mm.





Step 7: Sketch a line on the top face, from the front of the car 50mm long. Sketch another line cutting the full top face. Using the split body tool, select the main body and the horizontal line as the splitting tool. Tick the extend splitting tool option.





The Nose cone is finished.

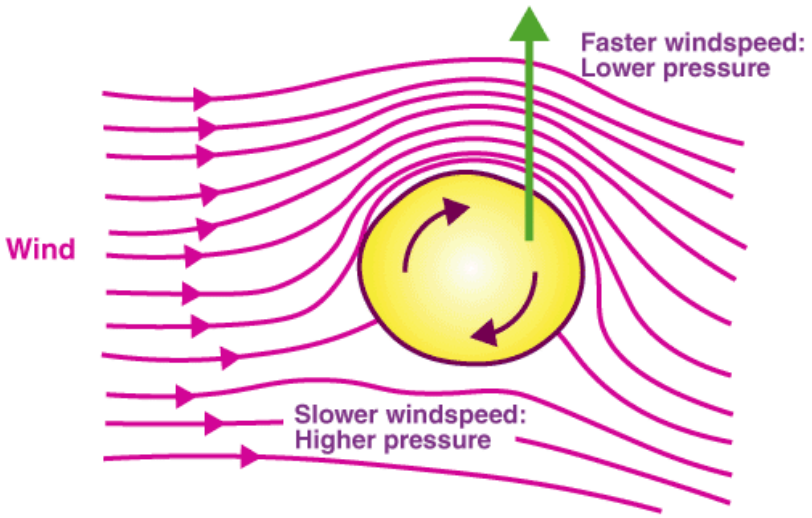
Front Wing:

The front wing is responsible for three things:

1. To cut the air going towards the wheels
2. To redirect air above the wheels
3. Using airfoil and different shapes, reduce drag

The main purpose behind the first two goals of the front wing is the magnus effect, which was discussed in Chapter 2.

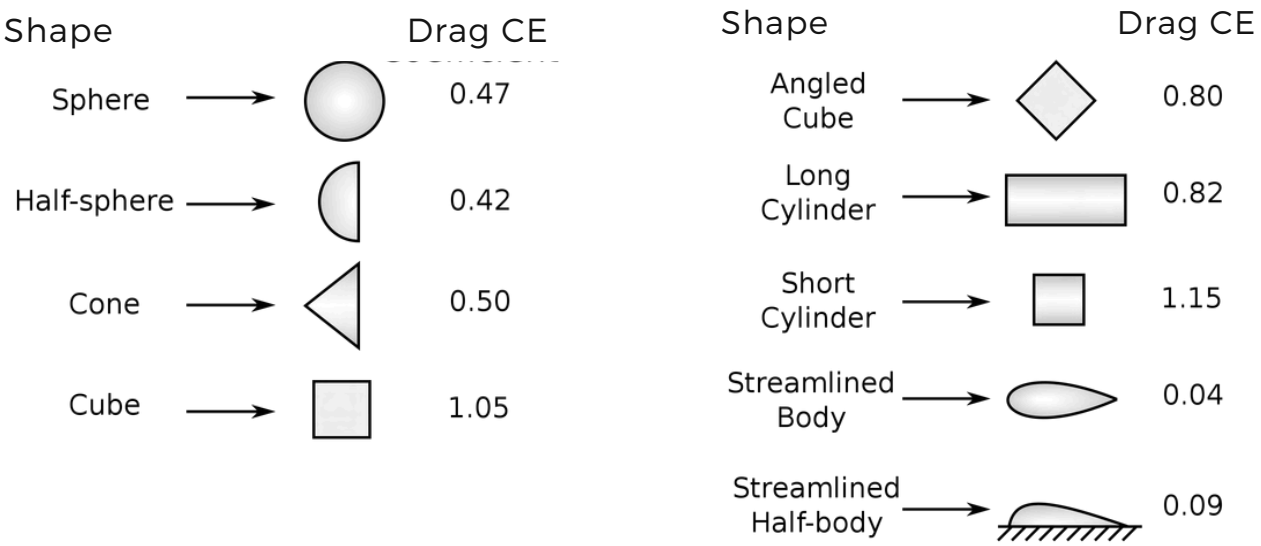
To summarise, the magnus effect occurs on a round object rolling in a direction. The air acts against the direction of movement, creating a topspin, slowing down the rotation, hence the speed of the car.



The air that hits the top part of the feel will create anti spin, since it forces the wheel to spin in the opposite direction. The air that hits the middle pushes the wheel, such as air ressistance against a flat object. The air that hits the bottom, will force a gap between the track and the wheel, creating lift.

An airfoil is a structure with curved surfaces designed to give the most favourable ratio of lift to drag in flight, used as the basic form of the wings, fins, and horizontal stabiliser.

To reduce the pressure due to air, we can cut the air in two or more parts, by creating a blade in the front wing.

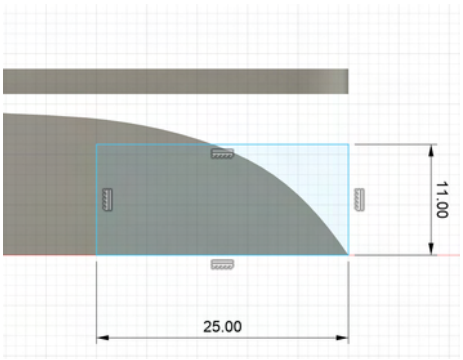


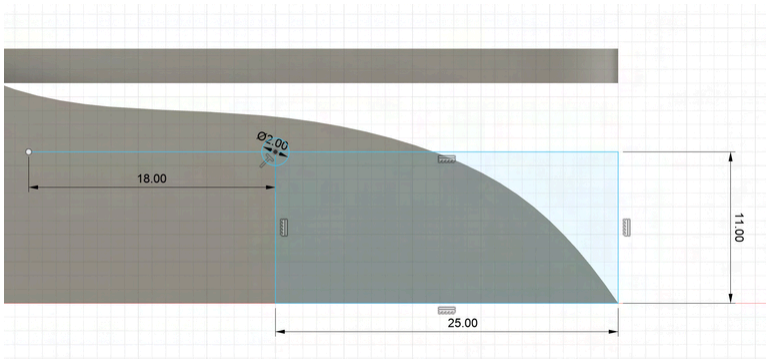
As we can see, a teardrop shaped airfoil creates the last possible drag coefficient when it comes in contact with air. Hence this is the shape we will use for our blade, which achieves the first and third goal of the front wing.

For the second goal, we have to create a curved slope so the air going toward the bottom part of the wheel is redirected above the car so it doesn't come in contact and disrupt the speed.

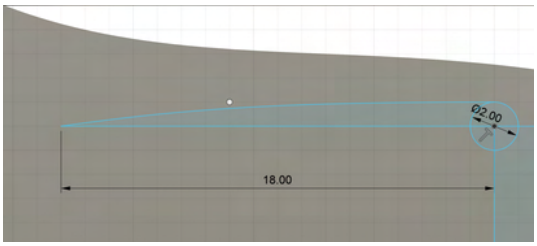
With the theoretical part of the front wing done, we can continue making it.

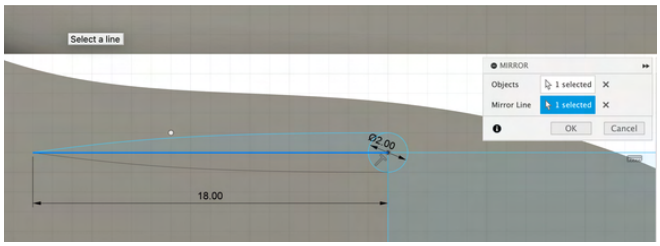
Step 1: Create a sketch on the side of the car. Draw a rectangle with height 11mm and length 25mm from the bottom right corner. Draw a circle with radius 2mm from the top left corner of the rectangle and a line of 18mm from the same point.



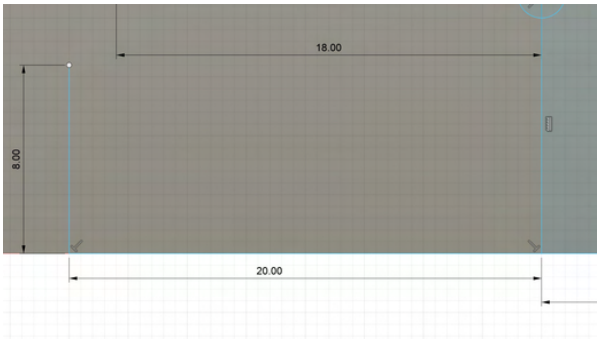


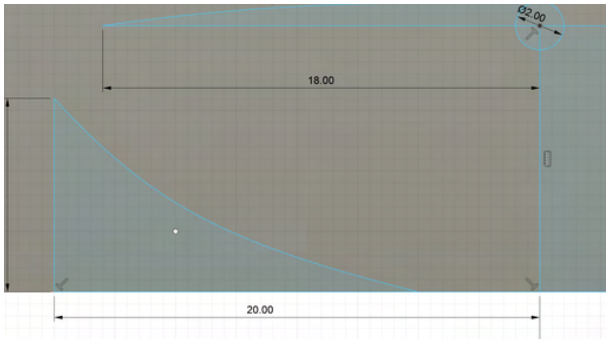
Step 2: Create a conic curve from the top of the circle to the other point on the 18mm line. Mirror the curve to the other side of the line.



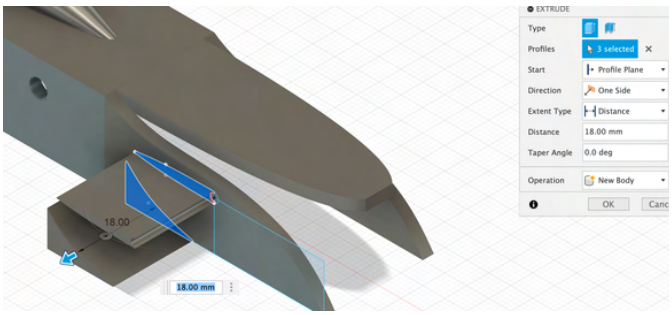


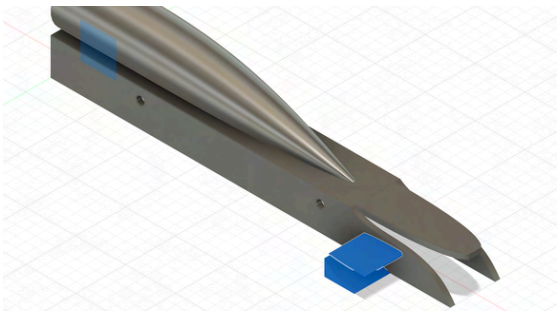
Step 3: Create a 20mm line from the bottom left corner of the rectangle and extend the height 8mm. Create a conic curve from the top of the 8mm line and 5mm away from the bottom left corner of the rectangle.



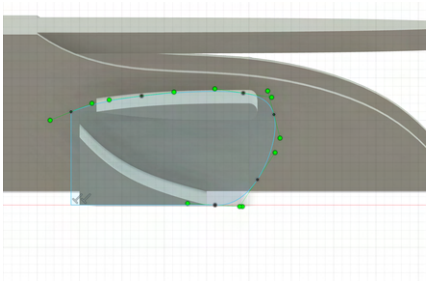


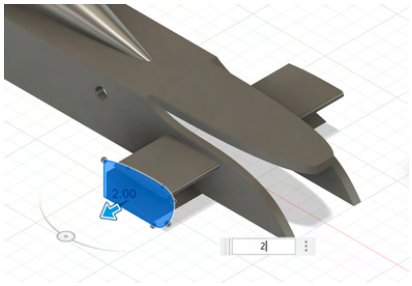
Step 4: Extrude the teardrop and the slope 18mm and select New Body instead of join. Mirror the bodies across the red plane

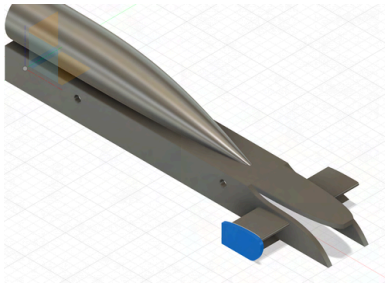




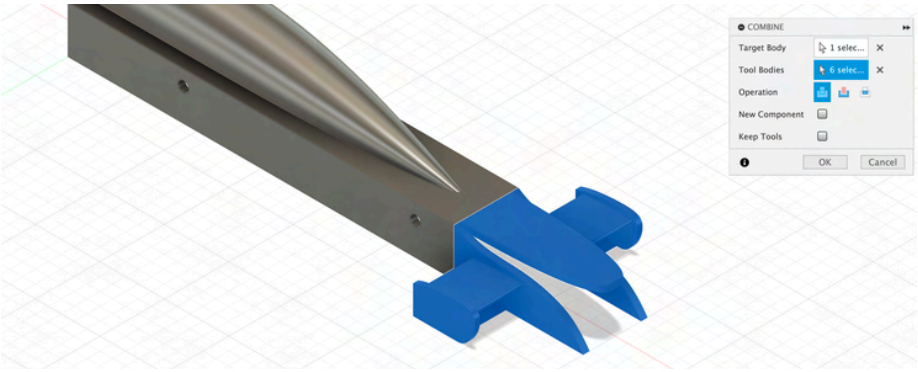
Step 5: Create a sketch on the side of the front wing. Using 2 lines and one filet, sketch the follwing outline around the wing and extrude it 2mm as a new body. Mirror along red plane.







Step 6: Use the combine feature to join all the wings and the endplates to the nose cone as the target body.



The front wing is finished.

Rear Wing

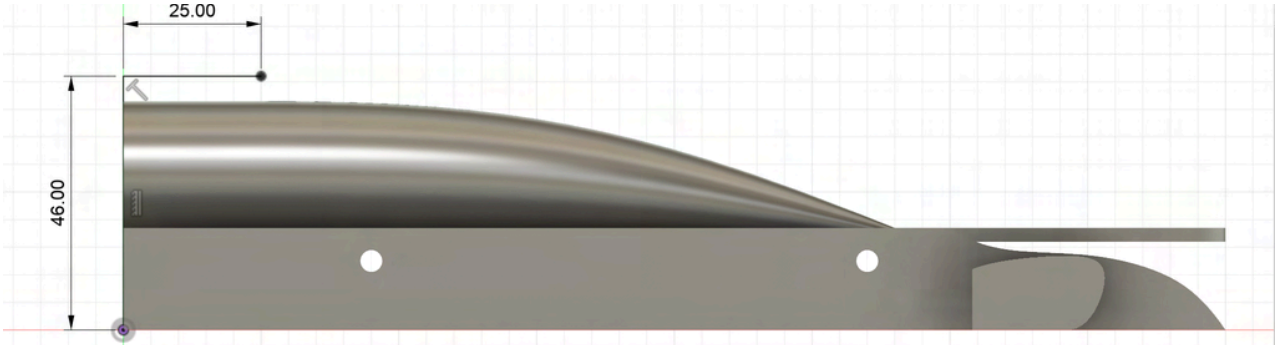
The most significant purpose of the rear wing is to generate downforce several times the weight of the car. This will push the car onto the track, improving its grip and speed.

The rear wing is positioned at the back of the car and creates a wake that flows over and around it, reducing the air resistance or drag.

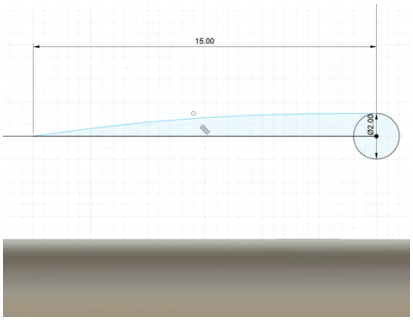
The front and rear wings produce the majority of the overall downforce, approximately 66%, on an F1 car. The rear wing balances out the downforce created at the front of the car to postition the car on the same vertical level throughout the race.

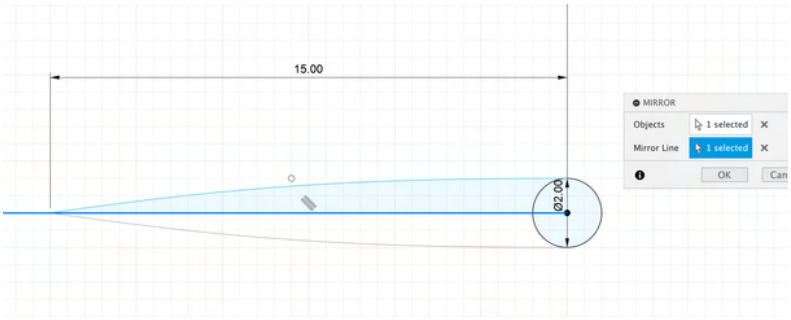
The same shapes analysed in the front wing development can be used here, which is the teardrop shape.

Step 1: Create a sketch on the red plane which is parallel to the side of the car. Draw a line from the bottom left corner of the car 46mm up and another line 25mm to the right.

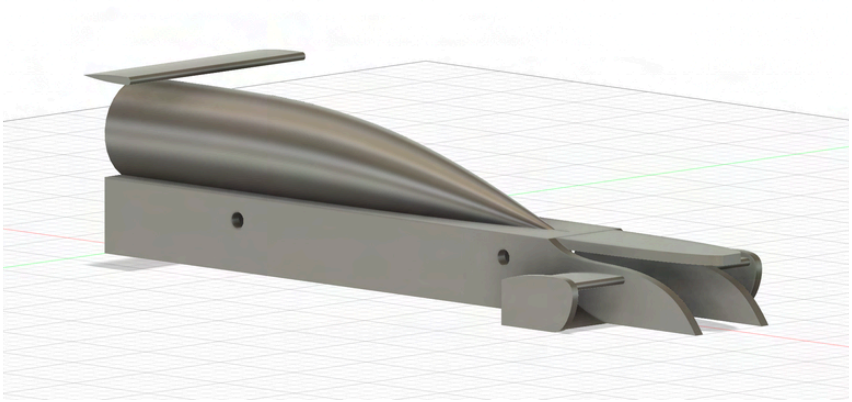


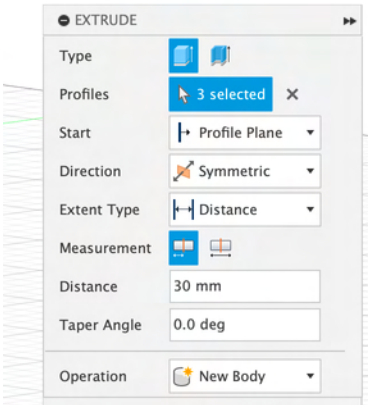
Step 2: draw a circle of diameter 2mm at the end of the 25mm line, and draw another line along the 25mm line of 15mm. Draw conic curves from the side of the circle to the end of the 15mm line and mirror it along the line.



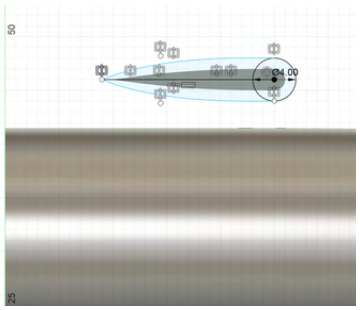


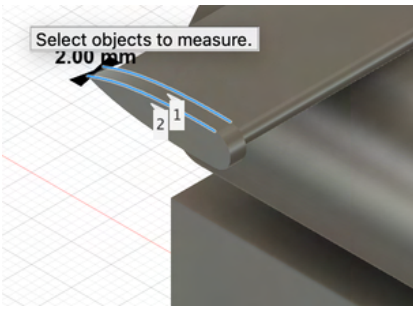
Step 3: Extrude the teardrop 30mm symmetrical on both sides.

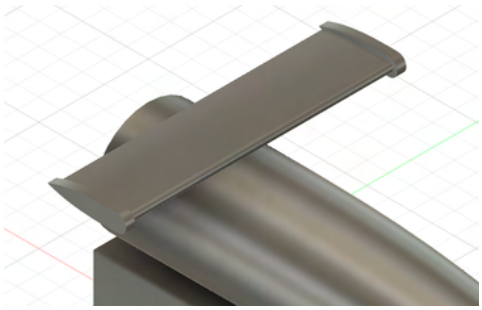




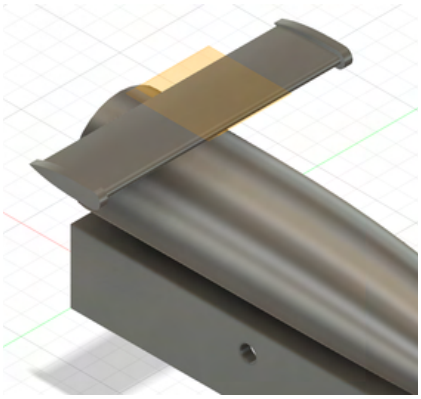
Step 4: Sketch a larger teardrop of the same shape on the side of the rear wing with a circle of 4mm and length of 16mm. Extrude it 2mm with as a new body and mirror to the other side of the wing.

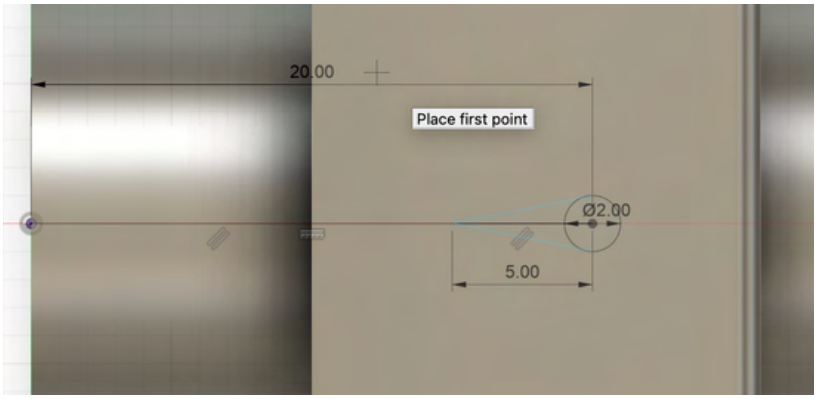




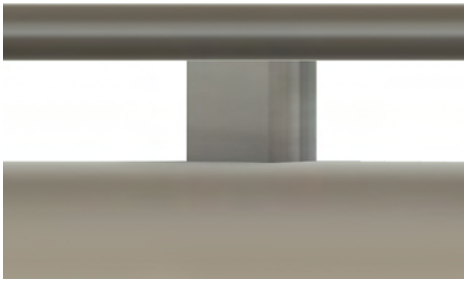


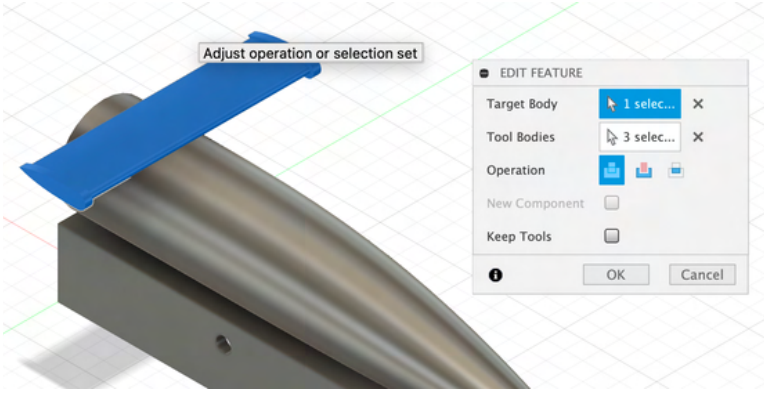
Step 5: Construct a plane facing up 41mm from the bottom.Create a line 20mm from the axis and a circle with diameter 2mm. Create another line 5 mm from the circle and connect the end of the line to the side of the circle with 2 lines.





Step 6: Extrude the teardrop 5mm upwards as a new body. Select all the parts created for the rear wing and join them using the join body command.





The rear wing is finished.

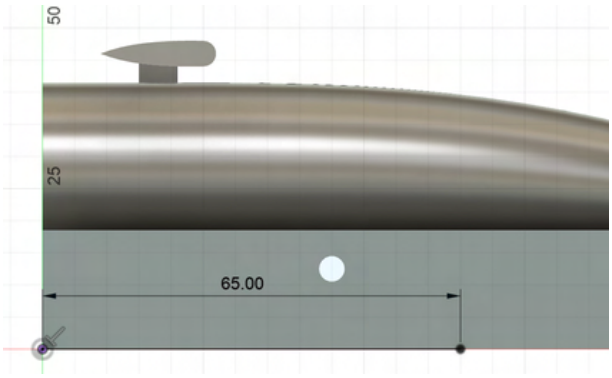
Side Pods:

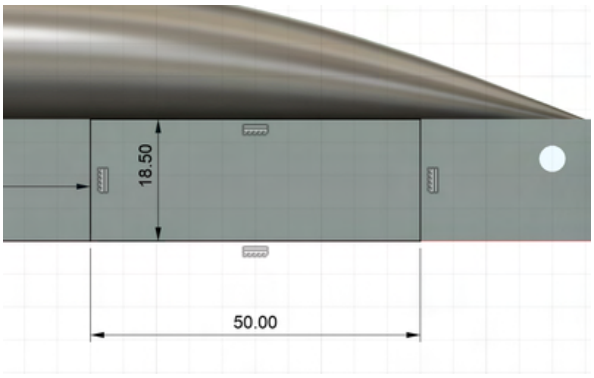
In Formula 1, side pods have multiple functions, such as engine cooling, housing fuel tanks and electronics, crash structures, and more. Although only one can be applied in this case.

Side pods play a crucial role in shaping the aerodynamic profile of an F1 car. They are designed to enhance the airflow around the vehicle, reducing drag and generating downforce. The shape and contour of the side pods are carefully engineered to minimize air resistance and maximize the car's overall performance.

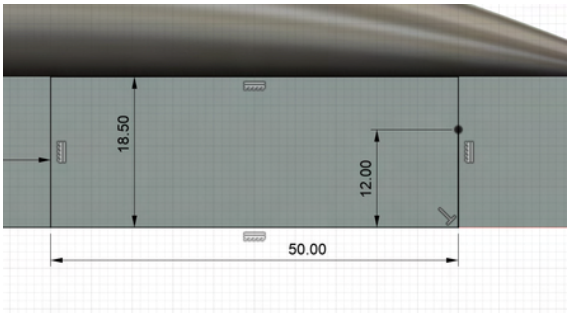
A lot of teams make the side pods hollow, and while this can be beneficial in reducing the mass of the car, it can prove to be disastrous. Since they are made of polyurethane foam, they are not very strong, and keeping the sides thin can cause it to break easily. Another reason to not keep side pods hollow, is to lower the centre of mass, to make the car more stable.

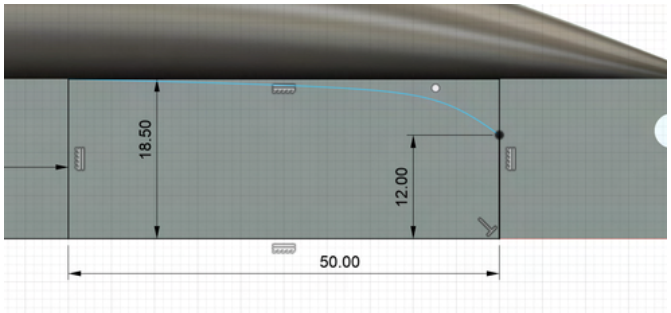
Step 1: Create a sketch on the side of the car. draw a line 65mm from the bottom left corner into the car. Create a rectangle of height 18.5mm and length 50mm.



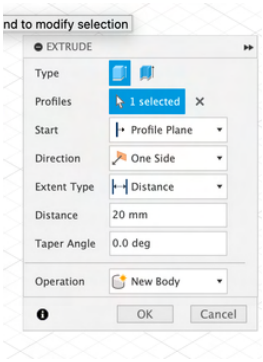


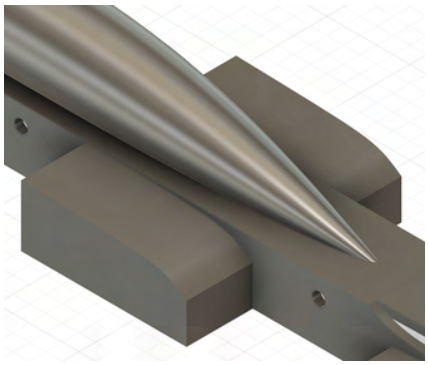
Step 2: From the bottom right side of the rectangle, create a line 12mm up and then a conic curve connecting that line and the top left corner of the rectangle as shown below.

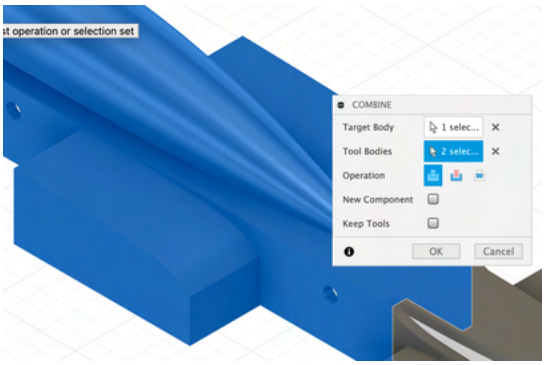




Step 3: Extrude the sketch 20mm to the side as a new body. Mirror the body to the other side of the car and use the combine tool to join both the side pods and the body.



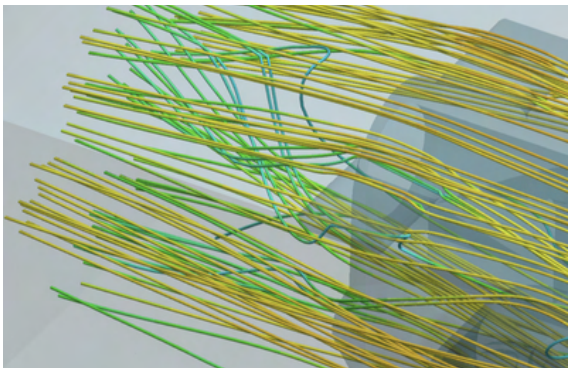




The side pods are finished.

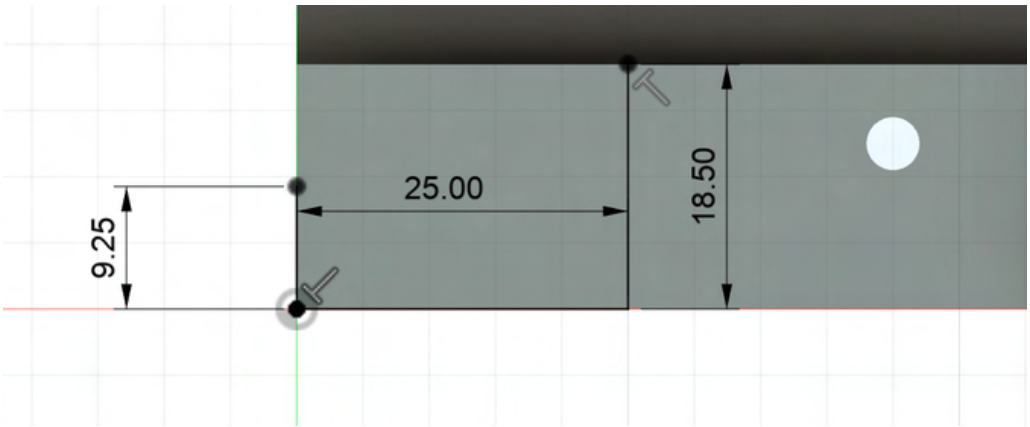
Rear Pods:

The rear pods can designed in a way to create a vortex like shape at the back of the car, which can be used to boost the car such as shown below.



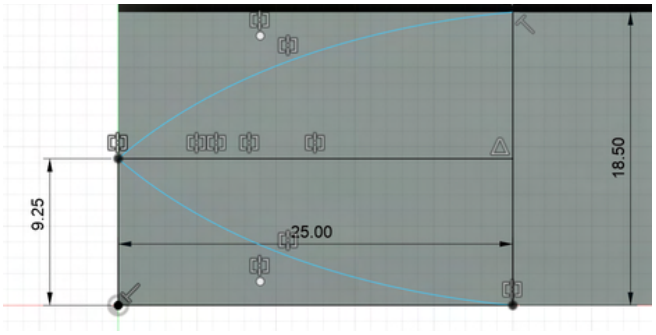
The design of the rear pods affects the car's balance and stability during high-speed maneuvers. Rear pods can be used strategically to optimize the weight distribution of the car. Proper weight distribution is crucial for achieving optimal handling characteristics.

Step 1: Create a sketch on the side of the car. Sketch a line 25mm from the bottom left corner and bring it up 18.5mm. From bottom left corner, sketch another line up 9.25mm.

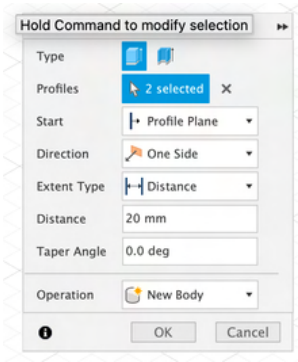


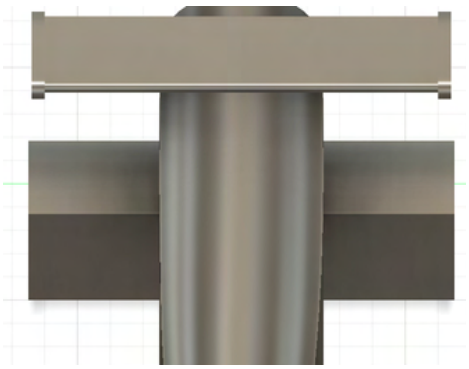
Step 2: Create a horizontal line from the 9.25mm line, which divides the 18.5mm line in 2. Draw a conic curve from a an end of the 18.5mm line to the top of the 9.25mm line. Mirror it to the other side of the horizontal line.

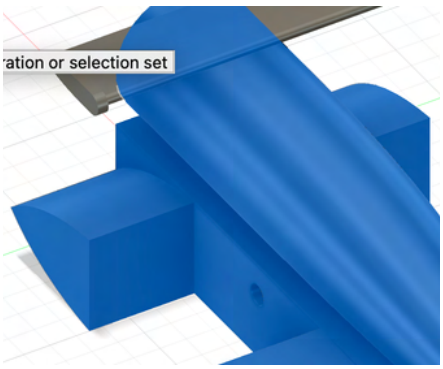




Step 3: Extrude the rear pods 20mm to the side as a new body, repeat the process of mirroring the pods to the other side and combining them with the body.





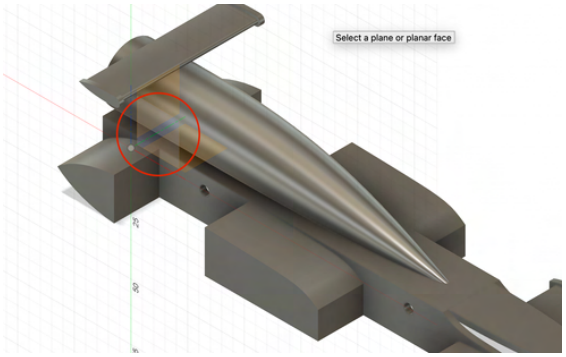


The rear pods are finished.

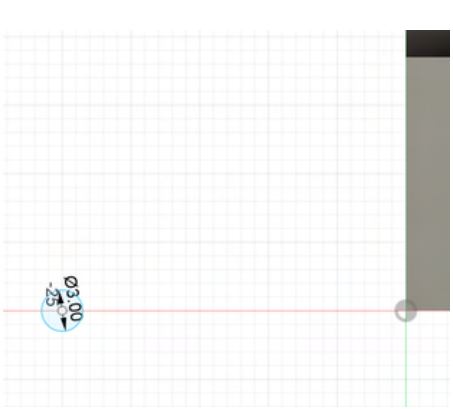
Wheels:

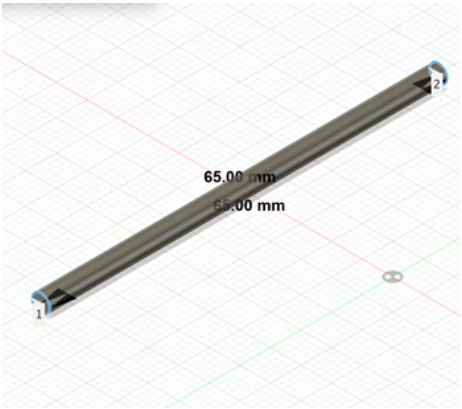
Wheels have a straightforward task, to carry the car. Hence they need to be strong enough to support the car but light enough to not weigh the car down. Wheels can also use ball bearings around the axils to smoothen the rolling by nullifying friction and increase the overall speed of the car.

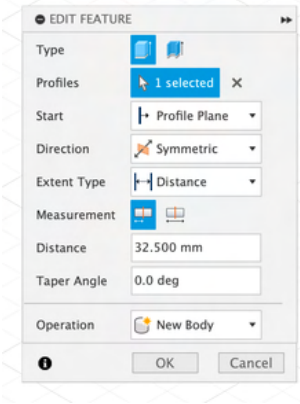
Step 1: Create a sketch on the plane which is parallel to the side of the car.



Step 2: create a circle of diameter 3mm outside the car. Extrude that circle 32.5mm symmetrically on both sides.

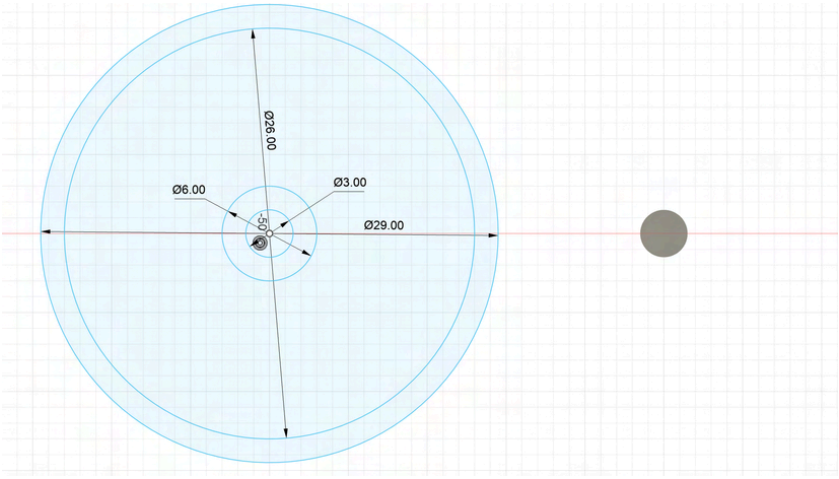




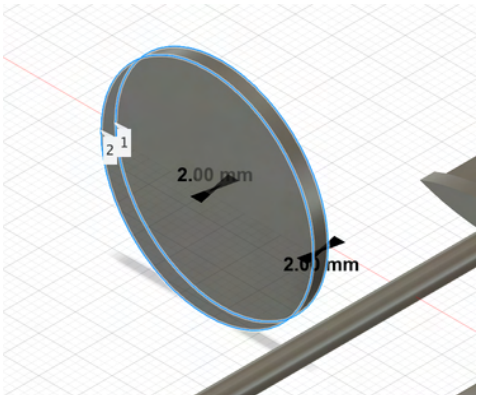


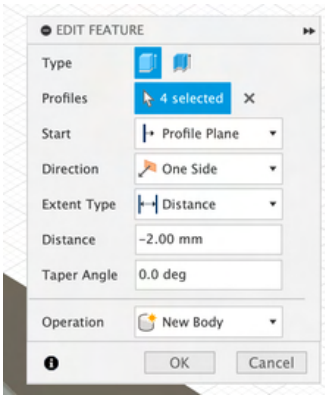
This is the axil of the wheels.

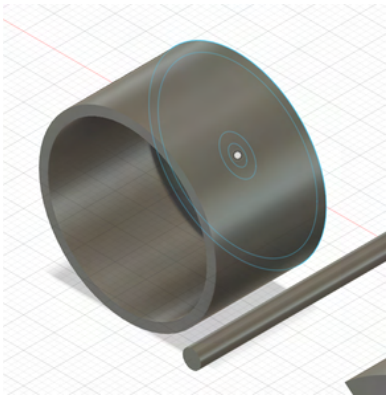
Step 3: Create a sketch on the same plane as earlier. Sketch 4 circles all from the same centre, of 29mm, 26mm, 6mm and 3mm respectively.



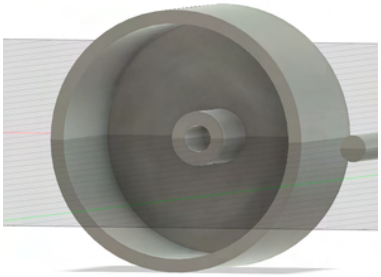
Step 4: Extrude the 29mm circle, inclusive of the other 3 circles, 2mm to the left. Extrude the 29-26mm ring 16mm more, 18mm in total.





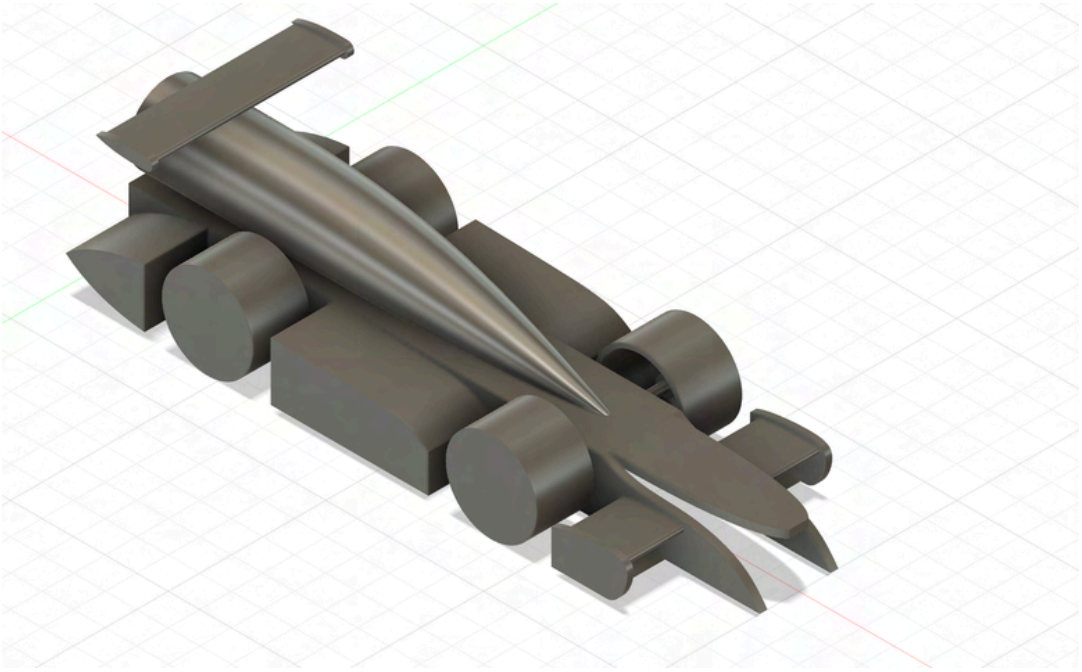


Step 5: Extrude the 6-3mm ring 8mm more to the left, making it 10mm in total. This will connect to the axil and hold it tightly.



Duplicate the wheel 4 times and the axil twice and align them with the axil holes of the car in a way that the axil goes into the body made in step 5, and keeps the wheel exactly 2mm away from the side of the car.

If you have done everything till this moment right, the car should look like this:



The finishing of this car can be done by adding fillets to all the edges of 0.5 to 1mm. Adding color and rendering it can be done using the Render workspace and appearance feature.

After all that is done, the car is ready.

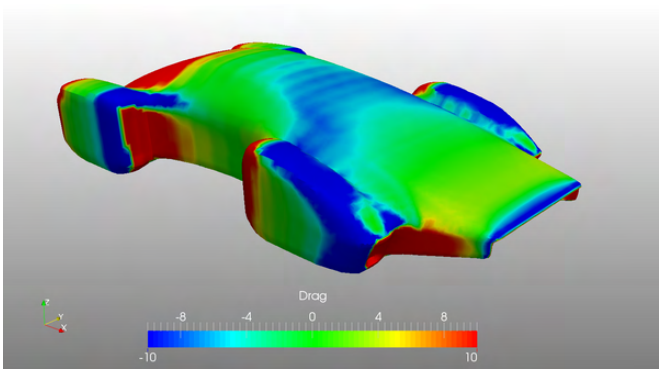
Analysing Simulations:

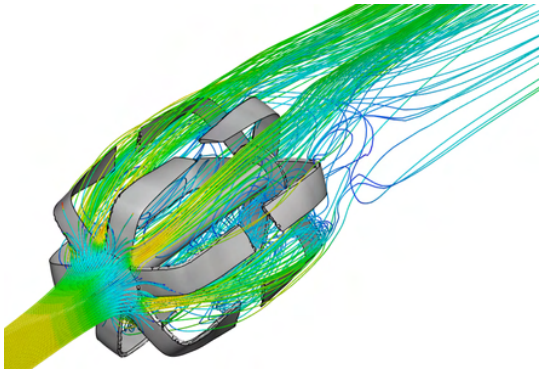
There are many softwares to analyse simulations and perform CAD such as Autodesk CFD, Ansys, Simscale, Air-shaper and more. However, all of these apps need a paid subscription so we suggest to do thorough research on which software would be best to test your car since every team has different requirements.

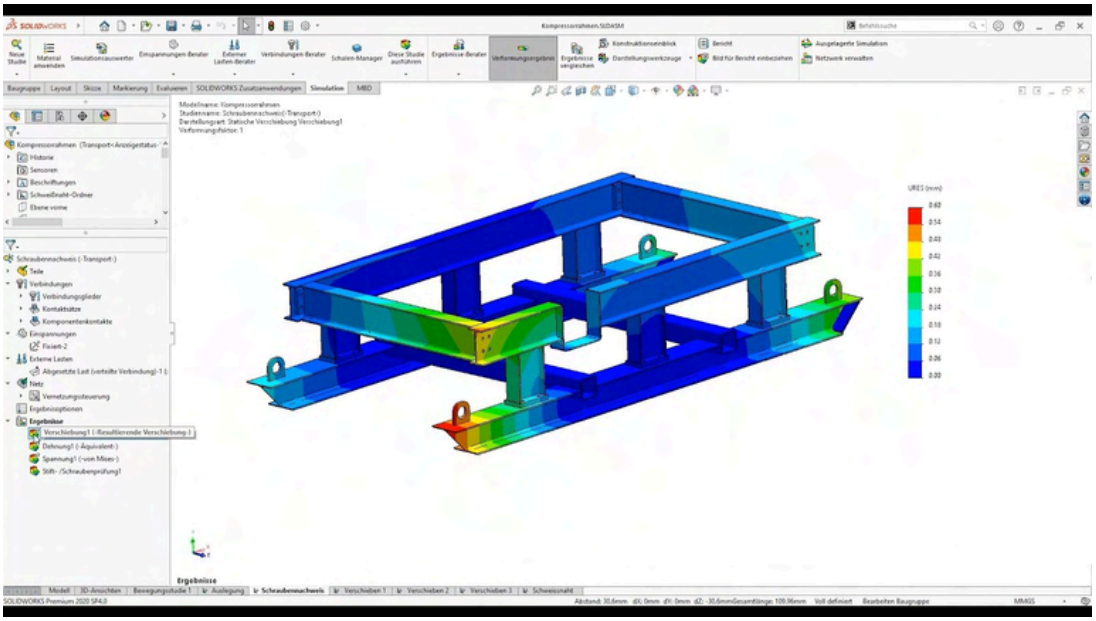


Simulations play an important role in the design and optimization process. There is a few tests that you should conduct to analyse your car and make improvements wherever needed. Here are the simulations needed:

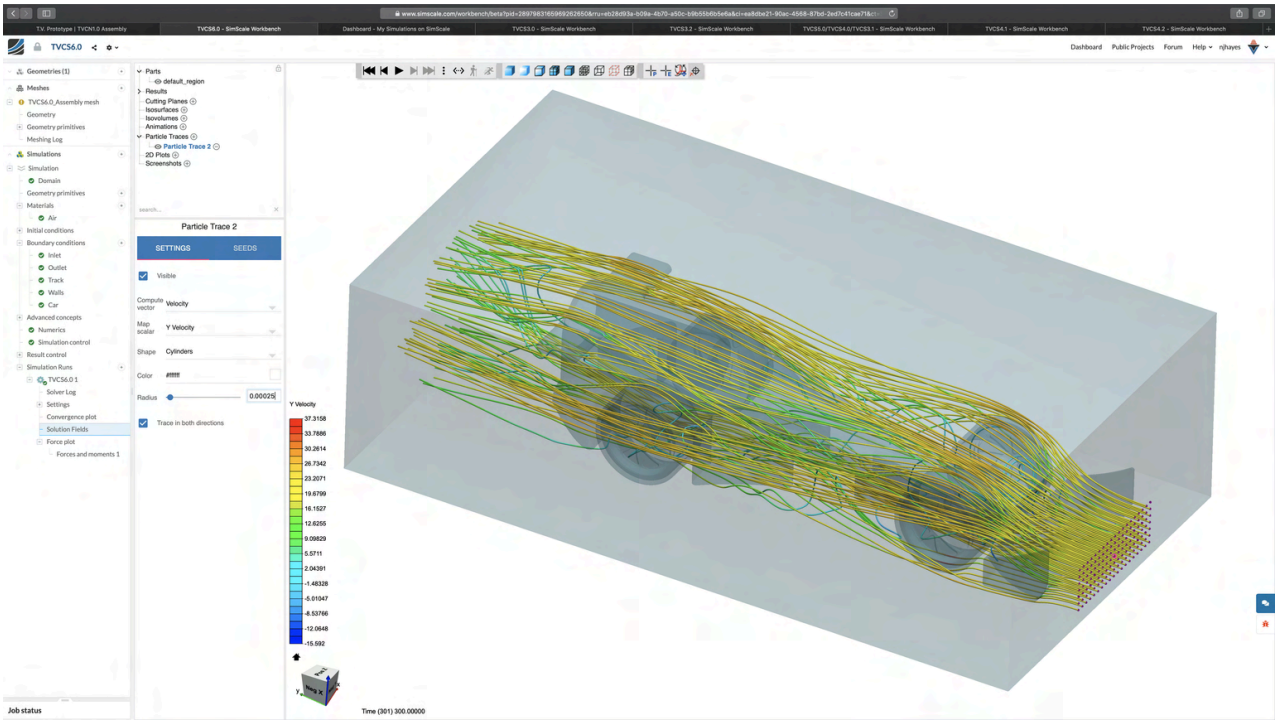
- Aerodynamics Simulation: This simulation helps teams analyze the airflow around the car, assess drag and downforce, and optimize the car's aerodynamic performance. It can be used to improve the design of the car's body, wings, and other aerodynamic components.
- Computational Fluid Dynamics (CFD): CFD simulations are used to study the fluid flow behavior around the car. This analysis helps teams understand how the air moves over and around the car, allowing them to make design changes that minimize drag and maximize downforce.
- Structural Analysis: This simulation involves evaluating the structural integrity and strength of the car's components, such as the chassis and suspension. It helps teams ensure that their car can withstand the forces and impacts experienced during races.







- Vehicle Dynamics Simulation: This simulation focuses on understanding the car's handling characteristics, including its acceleration, braking, and cornering behavior. It allows teams to optimize factors like weight distribution, suspension setup, and tire performance to improve the car's overall performance on the track.
- Virtual Wind Tunnel: Similar to the aerodynamics simulation, the virtual wind tunnel simulation allows teams to evaluate the impact of different design elements on the car's aerodynamic performance. It can provide insights into how changes in the car's shape, wing angles, or other features affect its overall performance.



These simulations are crucial for teams to fine-tune their designs, identify areas for improvement, and make data-driven decisions to optimize their cars' performance. It's important to note that the specific simulations may vary depending on the resources available to each team and the rules of the competition.

CHAPTER 5:

TEST YOURSELF

Now that you have learnt quite a lot on how to make an efficient and winning car design, keeping in mind the several scientific concepts that go behind the making of the car, we have prepared a fun quiz for you to try on.

Aerodynamics Quiz:

1. **Which of these is not a type of drag force?**

☐ Form Drag ☐ Skin Friction Drag ☐ Upthrust

2. **How does the lift force affect the car?**

☐ Pushes it up ☐ Pushes it down ☐ Pushes sidewards

3. **Which force produces a force opposite to lift?**

☐ Downforce ☐ Upthrust ☐ Friction

4. **The thrust force experienced by the car can be increased by decreasing _____ of the car?**

☐ Mass ☐ Size ☐ Material

5. Magnus effect is observed when a _____ object is moving through a fluid?

- [ ] Spinning
- [ ] Stationary
- [ ] Moving forward

6. Due to Coanda effect, the air _____ to convex surfaces?

- [ ] attracts
- [ ] sticks
- [ ] repels

7. Centre of mass can affect the _____ of the car?

- [ ] Stability
- [ ] Weight
- [ ] Size

8. Where should the centre of mass be located in the car?

- [ ] Lower part
- [ ] Middle
- [ ] Upper part

9. What is another term for centre of mass?

- [ ] Centre of force
- [ ] Centre of gravity
- [ ] Centre of body

10. Which motion does moment of inertia affect?

- [ ] Oscillatory
- [ ] Linear
- [ ] Rotational

11. What is the external force setting the F1 in schools cars in motion?

☐ CO_2 cannister force ☐ Drag ☐ Downforce

12. Understand Newton's second law can help increase the _____ of the car?

☐ Weight ☐ Acceleration ☐ Stability

13. When a forward force is produced by the car, there is a equal force in which direction?

☐ Forward ☐ Bakward ☐ Sideward

14. Which of the following types of friction, have significant impacts on F1 in schools cars?

☐ Kinetic Friction ☐ Static Friction ☐ Rolling Friction

15. What can be used in the wheels to reduce rolling friction?

☐ Axles ☐ Bearings ☐ Treads

For our last section of 'Test Yourself' go through the following statements and answer based on whether they are true or false. Circle the answer you think is right.

True or False?

The grip of the wheels with the ground is enhanced due to the effects of the lift force.

TRUE **FALSE**

The front wing of the car faces the most amount of downforce.

TRUE **FALSE**

The most important role of the rear wing of the car is to redirect the airflow around the car.

TRUE **FALSE**

The CO_2 cannister provides the thrust force for the car.

TRUE **FALSE**

Using high density 3D printing material for the manufacturing of the car , will increase its acceleration.

TRUE **FALSE**

Coanda effect is the tendency of the air to stay in contact with the surfaces of the car.

TRUE **FALSE**

The centre of mass of the car should be closer to the track to increase the stability of the car.

TRUE **FALSE**

The centre of mass is located at the centre of the car body.

TRUE **FALSE**

The thrust force received by all the F1 in schools cars is constant.

TRUE **FALSE**

There is an equal and opposite force for every force.

TRUE **FALSE**

The fluid friction which affects F1 in schools is the friction between the wheels and the track.

TRUE **FALSE**

Use of bearings in the wheels of the car can increase the rolling friction .

TRUE **FALSE**

REFLECTION

As a last task of this book, we'd like you to write a small reflection of what you have learned from this book and what you think you can do to design a better car. Write in the space provided below.

CHAPTER 5 :

SUPPORT US

We highly appreciate your purchase of this book. We hope you found it informative, fun and knowledgeable as well as learnt tons of designing a F1 in Schools car in basic.

As we embark on one of our most ambitious campaigns yet, we'd love each and every one of your support. Wish us all the best for the F1 In Schools World Finals 2023 as we represent our country at the highest and most challenging level of the competition. However that's not all!

By supporting us, we'd also like you to support the our many campaigns we hold as a team across our journey in Formula 1 In Schools. For further details on the campaigns we hold, view our Social Media pages (Instagram, Twitter, Facebook, YouTube). If possible, do connect with us in any way possible if you'd like to support us in any campaigns. Do message us in any support you need through our Infinity Junior campaign where we support newly participating teams to the challenge. Through our tutorials on YouTube, do not hesitate to send us images of cars you designed. We would love to see incoming messages from you.
For contacting us for any additional details:


@f1s.infinity


@f1Infinity_in

We hope to see absolutely fast cars by you on the race track after reading this book. Let us know if we can help you in any way possible.

Till then, Signing off,
Infinity Racing

INFINITY RACING

ESTD. 2022

AERODYNAMICS SIMPLIFIED

LAKSHYA HIRAWAT & ARYAMAN KEJRIWAL

India's F1 in Schools World Finalists Aryaman Kejriwal and Lakshya Hirawat wrote the book Aerodynamics Simplified. You will discover many things about designing a car through this book that you probably did not know previously. This book provides information on a wide range of topics relating to aerodynamics, the scientific concepts that go behind the making of a F1 in Schools car, motorsport, making a F1 in Schools car, and a host of other topics. If you are up to the challenge, there are also enjoyable activities scattered throughout the book to test your knowledge.

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F1 in Schools STEM Challenge